An Exponential Succinctness Gap between Three-Variable Logic and the Calculus of Relations

Yuya Uezato
CyberAgent, Inc.
uezato_yuya@cyberagent.co.jp

Yoshiki Nakamura
Chiba University
nakamura.yoshiki.ny@gmail.com

Abstract

Three-variable first-order logic and the calculus of relations define the same binary queries, but their classical translation can be exponential. We prove that this blow-up is unavoidable over finite structures, even when the target is a circuit with shared subterms and unrestricted complement and converse. Our source formulas are positive, have one quantifier, and have linear size over growing binary signatures. The proof uses a finite edge-colored reference graph with abundant witnesses: every intermediate relation on the reference input has a color table, yet removing one witness color at a vertex changes the query at an unchanged read edge. Transport reinterprets the reference tables on these recolored inputs. Boolean operations and converse commute with transport, while each composition can fail to commute on at most two recoloring tests. Counting these exceptions, with an additional input-occurrence argument for trees, gives circuit, tree-size, size-specific, and bounded-error randomized lower bounds. The same construction gives an exponential gap between three- and two-variable sentences, together with exact quantifier complexity and asymptotically optimal formula-size bounds for an explicit two-variable-definable query family. Extending the local calculation to finite join breadth yields exponential lower bounds for matrix query languages, even with arbitrary uniform entrywise operations and two-input aggregation kernels.

1 Introduction

Three-variable first-order logic (FO^3) over binary relations and the calculus of relations (CoR) define the same binary queries, an equivalence originating in Tarski's work in the 1940s and developed in Tarski and Givant's monograph [TG87]. The classical translation has a single-exponential size bound, but whether this increase is necessary for full CoR remained open [Nak22]. We prove that it is necessary over finite structures, even when the target shares intermediate results and uses arbitrary complement and converse.

The two formalisms organize intermediate information differently. A logical formula can keep three vertices in scope and combine conditions on the edges between them. A relational expression combines binary relations, eliminating an intermediate vertex whenever it composes two relations. This makes the size question relevant to compiling logical specifications into relational query plans and Boolean matrix programs: is an exponential increase inherent in the permitted operations, or merely an artifact of the translation?

For a structure with nonempty domain D, CoR uses the constants $\mathbf{0} = \varnothing$, $\mathbf{1} = D^2$, and $\mathbf{id} = \{(u,u) : u \in D\}$, together with union, intersection, complement $\overline{X} = D^2 \setminus X$, converse $X^\smile = \{(v,u) : (u,v) \in X\}$, and composition

$$X;Y = \{(u,v) : \text{some } w \in D \text{ satisfies } (u,w) \in X, (w,v) \in Y\}.$$

Composition implements the basic elimination step

$$\exists z\,(P(x,z) \wedge Q(z,y)) \quad \Longleftrightarrow \quad (P;Q)(x,y).$$

Its operands separately describe the edges entering and leaving the witness z. Conditions on the output pair (x,y) can be intersected with the result. The difficulty arises when Boolean choices couple the output edge to a witness edge: eliminating the witness may require separating all those choices.

1.1 Main results

Let $[k] = \{1,\dots,k\}$. Our lower bounds use the following positive formulas, for $k \geq 2$ and $k \geq 1$, respectively. Each has one quantifier:

$$F_k(x,y) = \exists z \bigwedge_{i=1}^{k} (R_i(x,y) \vee R_i(x,z)),$$

$$G_k(x,y) = \exists z \bigwedge_{i=1}^{k} [(P_i(x,y) \wedge Q_i(x,z)) \vee (Q_i(x,y) \wedge P_i(x,z))].$$

We call (x,y) the *read edge* and (x,z) the *witness edge*. For F_k, think of the R_i as requirements: a single witness edge must meet every requirement that the read edge misses. For G_k, the two edges make opposite choices of P_i and Q_i at each coordinate. On inputs with empty diagonals and exactly one of P_i, Q_i holding at each non-loop edge for each i, a non-loop read edge specifies a binary profile and a witness must have the complementary profile. We use such inputs for the lower bound, but both queries are defined on arbitrary inputs. The two families serve complementary purposes: F_k gives a particularly short specification, while G_k makes all 2^k profiles separately testable and yields sharp bounds for two-variable formulas.

A CoR term T is a syntax tree; $|T|$ counts its nodes and $\#_;(T)$ its compositions. A *relational circuit (DAG)* C allows shared subterms; $|C|$ counts distinct nodes and $g_;(C)$ distinct composition gates. Each input, constant, and operation node has unit size. Write $T \equiv_{\text{fin}} \varphi$, or $C \equiv_{\text{fin}} \varphi$, for equivalence on all finite structures over the input signature, and put $B_k = \binom{k}{\lfloor k/2 \rfloor}$.

Theorem 1.1 (Exponential lower bounds for full CoR)**.** *Every relational circuit C and* CoR *term T equivalent on finite structures to a query below satisfy the indicated lower bounds:*

Query	*Composition gates $g_;(C)$*	*Term size $\lvert T\rvert$*
F_k $(k \geq 2)$	$\lceil B_k/2 \rceil$	$2\lceil k/2 \rceil B_k - 1$
G_k $(k \geq 1)$	2^{k-1}	$2k2^k - 1$

The composition bounds also apply to terms.

Since $|F_k| = 4k$, $|G_k| = 8k$, and $B_k = \Theta(2^k/\sqrt{k})$, both families force $2^{\Omega(k)}$ compositions from a source of size $O(k)$. This remains true when all other operations are free and arbitrarily numerous. The signatures contain only binary relations and grow with k. For comparison, we record the classical upper bound; its proof is deferred to Appendix F.

Proposition 1.2 (Classical translation)**.** *Every* FO^3 *formula φ over binary relations with at most two free variables has an equivalent* CoR *term of size $2^{O(|\varphi|)}$ on all nonempty structures. Here size counts syntax-tree nodes.*

Thus a single exponential is both sufficient and necessary for worst-case FO^3-to-CoR compilation, even with sharing in the target.

Upper bounds and tightness. The corresponding CoR expressions are easy to write. For $I, A \subseteq [k]$, put $I^c = [k] \setminus I$, $A^c = [k] \setminus A$,

$$R_I = \bigcap_{i \in I} R_i, \qquad H_A = \bigcap_{i \in A} P_i \cap \bigcap_{i \in A^c} Q_i,$$

where an empty intersection is $\mathbf{1}$. Distributing the choices in the source formulas gives, on every nonempty structure,

$$F_k \equiv \bigcup_{I \subseteq [k]} \left[R_I \cap (R_{I^c}; \mathbf{1}) \right],$$

$$G_k \equiv \bigcup_{A \subseteq [k]} \left[H_A \cap (H_{A^c}; \mathbf{1}) \right].$$

Here a formula and a term are identified with the binary queries they define. The row projection $X; \mathbf{1}$ asks whether the first endpoint has an outgoing edge in X. In the first expansion, I lists requirements met by the read edge, and the projection asks for one witness meeting the rest. For example, when $k = 3$ and $I = \{1, 2\}$ the summand is $(R_1 \cap R_2) \cap (R_3; \mathbf{1})$. In the second expansion, A describes the choices at the read edge, and the projection asks for the complementary choices at one witness. These equivalences do not assume that P_i and Q_i are exclusive. They also give definitions in two-variable logic (FO^2), where the variable names can be reused; Section 1.4 quantifies the size required in that logic.

The all-read-edge case for F_k needs no composition, so its disjunctive normal form (DNF) uses at most $2^k - 1$ compositions. For G_k, the displayed DNF uses 2^k compositions. Sharing prefix intersections computes all H_A with $O(2^k)$ nodes, whereas writing the expansion as a tree uses $O(k2^k)$ nodes. Together with these expansions, Theorem 1.1 gives optimal circuit size $\Theta(2^k)$ and optimal term size $\Theta(k2^k)$ for G_k. The extra factor k for trees has a concrete origin: isolating a profile requires examining all k coordinates, and a tree pays again for input occurrences that a circuit can share.

For F_k, sharing improves even the DNF composition count. An implementation of Hansel's monotone identification method [Han66, AS17] gives

$$\lceil B_k/2 \rceil \leq \min_{C \equiv_{\mathrm{fin}} F_k} g_{;}(C) \leq B_{k+1} \leq 2B_k.$$

A witness meeting a set of requirements also meets every subset; the circuit exploits these implications between row projections. Appendix B implements the method with fixed wiring and $O(2^k)$ total nodes, determining the order of the optimal composition count. The tree-size bounds for F_k leave a gap between $\Omega(kB_k)$ and $O(k2^k)$.

What the bound measures. A logical formula with three variable names translates in linear size into relational algebra with at most three attributes per intermediate relation. Over a supplied domain relation, projection eliminates a quantified variable, and a Cartesian product with the domain restores that attribute when needed; the fixed set of three attribute names keeps the overhead per logical operation constant (Section 2). Our lower bound therefore exhibits an exponential cost of compiling such a three-attribute representation into the binary operations of CoR. Composition is a join followed by projection, so it applies to query plans built from these operations even when intermediate results are shared and Boolean conditions are unrestricted.

Equivalently, a relational circuit is a straight-line program over Boolean matrices: composition is Boolean matrix multiplication, converse is transpose, and the other instructions are entrywise Boolean operations and the zero, all-ones, and identity matrices. We bound the number of whole-matrix products in this model. Direct evaluation of G_k takes $O(kn^3)$ time on an n-element domain; our lower bound concerns the size and operations of the compiled program, rather than the scalar running time of arbitrary algorithms. Theorem 1.5 extends this arity-elimination result to the class of idempotent semirings specified there.

1.2 Natural approaches that fail

Fooling-set arguments. A natural first attempt at a lower bound for CoR is a fooling-set argument, the standard tool for lower bounds of this kind [LS81, KN97, AB09]. Such an argument is in fact the core of the only exponential lower bounds known so far, which concern two fragments of CoR [Nak22]: *positive* CoR, in which complement does not occur, and Σ_1CoR, in which it is applied only to input relations and **id**. In both, complement never stands above a composition; the two fragments correspond to existential positive and existential FO^3. For example, $\overline{R}; \mathbf{1}$ $(\exists z\, \neg R(x, z))$ lies in Σ_1CoR, but $\overline{R;\mathbf{1}}$ $(\forall z\, \neg R(x, z))$ does not. In [Nak22], the argument is phrased as a formula-size game, which hides its structure; recast as a fooling-set argument, it becomes transparent and shows exactly where complement breaks it. Since our approach is designed to overcome this failure, we spell the argument out.

The argument rests on two properties. First, composition is *rectangular*. It is the existential query $(X;Y)(x,y) \Leftrightarrow \exists z\,(X(x,z) \wedge Y(z,y))$, so a value w of z, called a *witness* for (u,v), must pass two separate tests: $(u,w) \in X$ on the left edge and $(w,v) \in Y$ on the right edge. Viewing relations as Boolean matrices, the pairs that lie in $X;Y$ through w therefore form a *rectangle* $\{u : (u,w) \in X\} \times \{v : (w,v) \in Y\}$: a set of rows times a set of columns, which need not be adjacent. Rectangles are coarse: if (u,v) and (u',v') lie in $X;Y$ *through the same witness* w, so do the crossed pairs (u,v') and (u',v). Covering many pairs using only rectangles, while avoiding some forbidden pairs, is therefore costly when the crossed pairs are forbidden, since no rectangle can then cover two of the pairs. This is the fooling-set argument of communication complexity, where a two-party function $f(a,b)$ is viewed as a matrix with rows a and columns b, the pairs to be covered are entries of value 1, and the forbidden pairs are entries of value 0. Second, Σ_1CoR is *monotone* under inclusion of structures. Write $\mathcal{M} \sqsubseteq_\iota \mathcal{M}'$ if ι maps $\mathcal{M}$ isomorphically onto an induced substructure of $\mathcal{M}'$, that is, $\mathcal{M} \subseteq \mathcal{M}'$ up to the renaming ι. Every Σ_1CoR term T satisfies, whenever $\mathcal{M} \sqsubseteq_\iota \mathcal{M}'$ and u, v are vertices of $\mathcal{M}$,

$$(u,v) \in T \text{ in } \mathcal{M} \quad \Longrightarrow \quad (\iota(u),\iota(v)) \in T \text{ in } \mathcal{M}',$$

since a composition asserts only that a witness exists, and ι maps witnesses to witnesses.

For each $k \geq 2$, [Nak22] constructs a query $\exists z\,\varphi$ of size $O(k)$, whose witnesses for a pair (u,v) in a structure $\mathcal{M}$ are, as for composition, the vertices w with $\mathcal{M} \models \varphi[x := u,\ y := v,\ z := w]$, together with structures $\mathcal{M}_1, \ldots, \mathcal{M}_t$, where $t = 2^k - 2$, and a structure $\mathcal{N}$ such that:

(i) each $\mathcal{M}_i$ has vertices u_i, v_i such that (u_i, v_i) has exactly one witness w_i;

(ii) $\mathcal{N}$ has vertices u^*, v^* such that (u^*, v^*) has no witness;

(iii) in both pairs, the two vertices are distinct and non-adjacent: no input relation contains (u_i, v_i) or (v_i, u_i) in $\mathcal{M}_i$, or (u^*, v^*) or (v^*, u^*) in $\mathcal{N}$;

(iv) for each pair $i \neq j$ up to order, $\mathcal{N}$ glues copies of $\mathcal{M}_i$ and $\mathcal{M}_j$ at their witnesses, so that the left edge (u_i, w_i) of $\mathcal{M}_i$ meets the right edge (w_j, v_j) of $\mathcal{M}_j$: some vertex c_{ij} satisfies $\mathcal{M}_i \sqsubseteq_\iota \mathcal{N}$ with $\iota(u_i) = u^*$, $\iota(w_i) = c_{ij}$, and $\mathcal{M}_j \sqsubseteq_{\iota'} \mathcal{N}$ with $\iota'(w_j) = c_{ij}$, $\iota'(v_j) = v^*$.

Rectangularity and monotonicity now let us run the fooling-set argument, with compositions as the rectangles, the structures $\mathcal{M}_i$ as the pairs to be covered, and $\mathcal{N}$ holding the forbidden crossed pairs. Let T be a Σ_1CoR term defining the query, and for $i = 1, \ldots, t$, call a subterm S of T *i-preserved* if $(u_i, v_i) \in S$ in $\mathcal{M}_i \Rightarrow (u^*, v^*) \in S$ in $\mathcal{N}$. By (iii), every *literal*, that is, a leaf (**0**, **1**, **id**, or an input relation) with any complement or converse applied to it, takes the same value on both pairs and is therefore i-preserved. By (i) and (ii), however, T is not i-preserved. Unions and intersections keep the property, so only a composition can break it: some composition $X_i;Y_i$ of T holds at (u_i, v_i) in $\mathcal{M}_i$ but not at (u^*, v^*) in $\mathcal{N}$, and a short additional argument [Nak22] lets it hold through w_i. We say that $X_i;Y_i$ *separates* $\mathcal{M}_i$ from $\mathcal{N}$. No composition $X;Y$ separates two structures $\mathcal{M}_i$ and $\mathcal{M}_j$ from $\mathcal{N}$. Otherwise, take c_{ij} from (iv), swapping i and j if needed. As $X;Y$ holds through w_i in $\mathcal{M}_i$ and through w_j in $\mathcal{M}_j$, we have $(u_i, w_i) \in X$ in $\mathcal{M}_i$ and $(w_j, v_j) \in Y$ in $\mathcal{M}_j$; monotonicity gives $(u^*, c_{ij}) \in X$ and $(c_{ij}, v^*) \in Y$ in $\mathcal{N}$, and rectangularity gives $(u^*, v^*) \in X;Y$ in $\mathcal{N}$, a contradiction. Each $\mathcal{M}_i$ thus needs a composition of its own, and a Σ_1CoR term defining the query cannot get by with fewer than $t = 2^k - 2$ compositions, although the query itself has size only $O(k)$.

In full CoR, complement destroys monotonicity, and the argument collapses entirely. Monotonicity was the indispensable step: it lets the tests passed in $\mathcal{M}_i$ and $\mathcal{M}_j$ be combined in $\mathcal{N}$. Just as complement reverses $\subseteq$, it reverses this implication: $\overline{R;\mathbf{1}}$ holds at (u, v) when u has no R-successor, but fails at $(\iota(u), \iota(v))$ once $\iota(u)$ gains one in a larger $\mathcal{M}'$. No choice of structures repairs this, since the argument must carry tests from the $\mathcal{M}_i$ into a structure that contains them. On the instances of [Nak22], the failure is complete: u_i has a single neighbor in $\mathcal{M}_i$, whereas u^* has many in $\mathcal{N}$, so a single term saying that u has at most one neighbor, which needs complement above a composition, separates every $\mathcal{M}_i$ from $\mathcal{N}$ at once, without looking at any witness.

Infinite, but not finite, counter-models. Proving a lower bound means finding, for every circuit with few compositions, a counter-model: a structure on which the circuit and the source formula disagree. If infinite counter-models are allowed, the task becomes much easier, because symmetry can then control every intermediate relation. Fraïssé's construction [Fra54] is the standard tool of model theory for building such symmetric structures [Hod93], and it is the natural first thing to try. Finite edge-colored complete graphs form a Fraïssé class: two of them can be amalgamated over a common subgraph by coloring the new pairs arbitrarily. By Fraïssé's theorem, this class has a countable homogeneous limit: an infinite edge-colored complete graph into which every finite one embeds and in which every isomorphism between finite subgraphs extends to an automorphism. Hence every pair of distinct vertices has infinitely many witnesses for every pair of colors, and every relation definable in CoR, being invariant under automorphisms, is a union of colors and possibly the diagonal. The limit is therefore a reference input of exactly the kind the argument below needs, and recoloring it at one vertex yields the tests from which counter-models are drawn. With this limit in place of a finite structure, Lemma 3.1 becomes unnecessary and Lemma 3.2 follows from homogeneity; the rest of the argument (Lemma 3.3, Theorem 3.4, and Section 4) applies unchanged and gives the lower bound for translations required to be correct on all structures.

This route fails for the finite structures that computation requires. A query plan or a matrix program needs to be correct only on finite inputs, and a lower bound witnessed by infinite structures does not exclude a short translation that is correct on finite ones. Finite counter-models cannot simply be cut out of the limit. Already for graphs, finite homogeneous structures are rare [Gar76], so a finite piece loses homogeneity, and the limit does not reveal which finite pieces keep its extension properties. Such finite structures come from separate probabilistic or algebraic constructions [BEH81], and for closely related limits their existence is open: for the generic triangle-free graph, it is unknown whether finite graphs with the four-point extension property exist [Che11, LS20].

1.3 Technical overview of Theorem 1.1

Our technique: equality on local changes and random reference graphs. Our approach avoids both obstacles. Complement reverses inclusion but preserves equality, so we compare computations by equality. For this, the inputs we compare share their domain and coincide except at a few edges, which also removes the global features that defeat fooling sets. We fix one input and t changes of it, each of which flips the query at a pair where the input itself is unchanged. A correct circuit must notice every change, and since Boolean operations and converse act entry by entry, only compositions can carry a change from elsewhere to that pair. This alone does not force t compositions: for $R = \{(h, w_i)\}_{i \le t}$ and $S = \{(w_i, v_i)\}_{i \le t}$, deleting (h, w_i) removes (h, v_i) from $R; S$, so this single composition notices all t deletions. Our input is therefore chosen so that each

composition notices at most two changes, and at least $t/2$ compositions are needed.

The input is a finite edge-colored complete graph chosen at random. Unlike a homogeneous structure, it needs only a weak extension property: for all distinct vertices u, v and all colors λ, μ, two vertices z outside $\{u, v\}$ satisfy $\chi(u, z) = \lambda$ and $\chi(z, v) = \mu$. Coloring the edges independently gives this property with $O(N^2 \log N)$ vertices for N colors (Lemma 3.1), as in classical finite representations of relation algebras [JMT95]. This weak property already makes every intermediate relation a color table (Lemma 3.2), and the second witness lets a change at one vertex be avoided everywhere else. Since the construction works at every size $n \geq n_0(N)$, it also gives the fixed-size bounds of Corollary 1.3.

The rest of this subsection makes this precise for G_k. We describe the reference graph and the tests (Section 1.3.1), transport of color tables (Section 1.3.2), the two-color bound for compositions (Section 1.3.3), and the resulting count (Section 1.3.4). We then turn to the tests supplied by the source queries (Section 1.3.5) and to other targets (Section 1.3.6).

1.3.1 The reference graph and one-color folds

All 2^k tests use the same reference input. A test changes that input at one vertex and reads the query at an unchanged edge. Controlling arbitrary intermediate relations requires more than the existence of the query's own witnesses: the reference input must realize every pair of colors that a composition might use.

Use the palette $\Lambda = 2^{[k]}$, and write $\chi(u, v)$ for the color of the undirected edge $\{u, v\}$. Color one complete graph so that, for any distinct vertices u, v and colors λ, μ, at least two vertices z outside $\{u, v\}$ satisfy $\chi(u, z) = \lambda$ and $\chi(z, v) = \mu$. Thus every color pair has a witness even when one additional vertex is forbidden. This is a two-witness form of the classical random and affine-plane constructions of Jipsen, Maddux, and Tuza [JMT95]; independent coloring and a union bound give $O(|\Lambda|^2 \log |\Lambda|)$ vertices. Section 3.1 proves the exact-size statement we use.

On a color-μ edge, let P_i hold for $i \in \mu$ and Q_i for $i \notin \mu$. Exactly one of P_i, Q_i is therefore true for each coordinate on each non-loop edge; both are false on loops. Fix a vertex h, the *hub*. For each $A \in \Lambda$, recolor every A-colored edge at h to A^c. This *one-color fold* removes exactly color A from the hub. Read the query at (h, v_A), an edge of color A^c that the fold leaves unchanged. The circuit receives only the input relations, without constants naming the hub or the read vertices.

For example, take $k = 2$ and $A = \{1\}$. The read edge has color $\{2\}$, so its true predicates are Q_1, P_2. The clauses of G_2 require a witness with P_1, Q_2: exactly color $\{1\}$. Such witnesses exist before the fold, and all disappear afterwards. A loop cannot replace them because its input predicates are false. The query changes even though every input relation is unchanged at the read edge (Figure 1).

1.3.2 Color tables and transport

The query now distinguishes the reference input from each folded input. To turn these distinctions into a lower bound, we need a description of every value a circuit can compute on the reference input. A relation has a *color table* if its value is one bit on the diagonal and one bit per edge color. The input relations and constants have such tables. Boolean operations act entrywise, and converse preserves the tables because the coloring is undirected. Composition is closed as well. For distinct endpoints, the rich coloring realizes every pair of colors on the two edges through a witness distinct

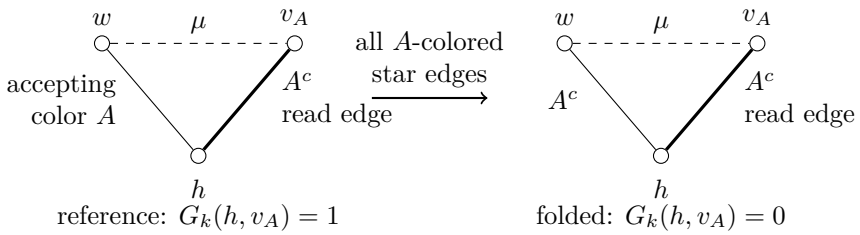


Figure 1: The only accepting witness color is A. The fold recolors all A-colored edges at h, of which one is displayed. The thick read edge and every edge away from h remain unchanged; μ denotes the color between the displayed non-hub vertices.

from both endpoints, which we call a *generic witness*. Existence therefore depends only on the sets of colors accepted by the operands. Equal endpoints admit every same-color pair. Witnesses equal to an endpoint contribute terms determined by the diagonal bits and the output edge color. Consequently *every* reference intermediate has a color table, however much complement or sharing is used. These tables form the classical finite relation algebra $E_{|\Lambda|+1}(1,2,3)$ [JMT95]; our proof uses its multiplication table rather than any symmetry of a particular graph.

More generally, let $f : \Lambda \to \Lambda$ recolor just the edges at h. For a reference relation X, its *transport* $\mathsf{T}_f X$ is the relation obtained by reading the same table on the recolored graph. This map preserves Boolean operations, including complement, as well as converse and the constants. It also preserves values at unchanged read edges. At each gate we therefore have two relations to compare: the *transported reference value*, obtained from the computation on the reference input, and the actual value obtained by running the circuit on the folded input. The inputs agree with their transported reference values by construction. Whenever the operands do as well, Boolean operations and converse propagate this agreement. Only composition can break it. This comparison does not assume that arbitrary actual values on the folded input still have color tables.

1.3.3 Two colors per composition

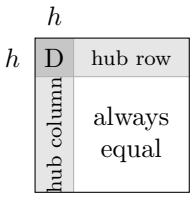
We next bound how many tests can break this agreement at one composition, regardless of how its operands were computed. Fix reference operands X, Y with off-diagonal table entries p_λ, q_λ. For $J \subseteq \Lambda$, define the Boolean joins

$$j_J^{\mathrm{L}} = \bigvee_{\lambda \in J} p_\lambda, \qquad j_J^{\mathrm{R}} = \bigvee_{\lambda \in J} q_\lambda, \qquad j_J^{\mathrm{D}} = \bigvee_{\lambda \in J} (p_\lambda \wedge q_\lambda).$$

Here an empty join is zero; L and R refer to the left and right operands, and D to their common support. Put $J = f(\Lambda)$, the colors still present at the hub. Away from h, the second graph witness allows us to avoid h, so recoloring a witness through h does not change the generic contribution. Contributions from witnesses equal to an endpoint commute with transport directly. The remaining comparison has just three cases, shown in Figure 2.

To preserve all three joins, if $j_\Lambda^{\mathrm{D}} = 1$, select one color with $p_\lambda = q_\lambda = 1$. It certifies all three. Otherwise, select one color for each of $j_\Lambda^{\mathrm{L}}, j_\Lambda^{\mathrm{R}}$ that is true. In either case this gives a set $K(p,q)$ of at most two colors such that

$$K(p,q) \subseteq f(\Lambda) \quad \Longrightarrow \quad (\mathsf{T}_f X); (\mathsf{T}_f Y) = \mathsf{T}_f (X; Y).$$



Output pair	$(\mathsf{T}_f X);(\mathsf{T}_f Y)$	$\mathsf{T}_f(X;Y)$
Hub row: (h,v), $v \neq h$	$j_J^{\mathrm{L}} \wedge j_\Lambda^{\mathrm{R}}$	$j_\Lambda^{\mathrm{L}} \wedge j_\Lambda^{\mathrm{R}}$
Hub column: (u,h), $u \neq h$	$j_\Lambda^{\mathrm{L}} \wedge j_J^{\mathrm{R}}$	$j_\Lambda^{\mathrm{L}} \wedge j_\Lambda^{\mathrm{R}}$
Diagonal: (h,h)	j_J^{D}	j_Λ^{D}

Figure 2: Possible failures of commutation with transport. The products agree away from the shaded hub row and column. The table compares their *generic witness contributions*; endpoint contributions agree and are omitted. Preserving the three joins preserves the complete product. The schematic displays h first only for illustration; no vertex order is available to the circuit.

The set is chosen from the fixed reference operands *before the recoloring map is chosen*, and works for every such map. A one-color fold can therefore be exceptional only when it removes a color in $K(p,q)$. This uniformity is what bounds the number of exceptional tests. The two colors in the certificate have a different role from the two graph witnesses: they support the three Boolean joins, while the graph witnesses allow a forbidden vertex to be avoided. If one operand is constant across off-diagonal colors, one supporting color suffices; if both are, no color is needed.

1.3.4 The test–cost theorem

The local bound applies to every gate of a circuit, including a gate whose output is used many times. Evaluate a circuit C once on the reference input. For a composition gate g with reference operands X_g, Y_g, let

$$D_g = \{A \subseteq [k] : (\mathsf{T}_A X_g);(\mathsf{T}_A Y_g) \neq \mathsf{T}_A(X_g;Y_g)\},$$

where T_A denotes transport for the fold removing A. For every other gate, put $D_g = \varnothing$. Thus $|D_g| \leq 2$. These sets count failures to commute with transport, not ordinary changes of a gate's value. In particular, their definition uses transported *reference* operands, even if earlier gates fail on the same test.

If A lies outside every D_g, topological induction shows that every actual gate value on the folded input equals its transported reference value. The circuit then gives the same answer at the unchanged read edge on both inputs, although G_k has different answers. Hence a correct circuit must cover all tests by the sets D_g. With $t = 2^k$ and the *canonical cost* $\mathrm{cost}(C) = \sum_{g \text{ distinct}} |D_g|$,

$$t \leq \left|\bigcup_g D_g\right| \leq \mathrm{cost}(C) \leq 2g_{;}(C).$$

A shared gate is counted once, regardless of its fan-out. This proves the circuit bound without converting the circuit to a tree.

The same accounting also measures the errors of an incorrect circuit. Each uncovered test forces an error on at least one of its two observations. Applying this statement to each random circuit and taking expectations gives $(1-2\varepsilon)t \leq \mathbb{E}\ \mathrm{cost}(C)$ when every observation has error probability at most ε. No independence between tests is needed. This is the test–cost theorem of Section 3.4.

1.3.5 Tests from the source queries

The argument separates the number of tests supplied by the source query from the cost of operations in the target. For G_k, every profile has one complementary witness profile, giving $t = 2^k$. For F_k, the acceptable witness profiles overlap, so we first choose a palette on which each test again has a unique accepting color.

Interpret R_i as true on color μ exactly when $i \in \mu$, and false on loops. A color-A^c read edge then accepts precisely the witness colors containing A. Restrict the palette to sets of size $k/2$ for even k, or to sets of sizes $\lfloor k/2 \rfloor$ and $\lceil k/2 \rceil$ for odd k. Choose target colors A of size $\lceil k/2 \rceil$. No target has a strict superset in this palette, so it again has a unique accepting color. The color A^c remains in the palette and is fixed by the fold from A to A^c. We therefore obtain $t = \binom{k}{\lfloor k/2 \rfloor} = B_k$ tests using exactly the same transport argument.

1.3.6 Other targets: formulas, trees, and matrices

The target language determines the local costs. Existential or universal quantification in FO^2 has cost at most one: it is a row or column projection, possibly combined with complement. For scalar aggregation, the same three positions in Figure 2 control the calculation. If every finite join equals a join of at most β of its summands, the scalar domain has *finite join breadth* β. Choosing such a support for each of the three contributions uses at most 3β colors, even when the function combining the two scalar operands is arbitrary. The corresponding bounds are:

Operation	Canonical cost at most
Uniform entrywise operation, converse, transpose, permitted constant	0
CoR projection or FO^2 quantifier	1
CoR composition	2
Matrix–vector or vector–matrix contraction	β
Square matrix–matrix contraction	3β

For Boolean scalars, the last bound improves to two even for arbitrary two-input kernels. The permitted diagonal, broadcasting, and vector inner-product operations have cost zero. Section 6 specifies the operations and proves reference closure for every array type. These costs measure failures of preservation, not evaluation time.

The additional charge for trees. Local operation costs already give lower bounds for circuits and hence for trees. The stronger tree-size result needs one additional argument. Selecting one target color requires k input occurrences on the cube and $\lceil k/2 \rceil$ on the middle-layer palette. A tree charging lemma assigns disjoint input occurrences to the operations that obstruct tests. This gives the extra factor in the relation-atom lower bound; it does not follow just by viewing a tree as a circuit.

1.4 Further results

The construction and its transport comparison also give bounds for size-specific and randomized computation, two-variable logic, and matrix query languages. We state these consequences here, together with an elementary independent separation from CoR to FO^2.

Fixed domain sizes and bounded error. For a palette of $N \geq 2$ colors, put

$$n_0(N) = \left\lceil 12N^2 \ln(2N) \right\rceil + 3 = O(N^2 \log N).$$

Coloring edges independently supplies our graph separately at each $n \geq n_0(N)$, and all its tests use the same domain. This allows a different target circuit to be chosen for each domain size. A randomized target is an input-independent distribution of circuits. Its error is *pointwise*: for each input and each output pair, the probability of a wrong answer is at most ε.

Corollary 1.3 (Fixed sizes and bounded error)**.** *Fix $k \geq 1$, $n \geq n_0(2^k)$, and $0 \leq \varepsilon < 1/2$. Every input-independent distribution of relational circuits computing G_k on all n-element inputs with pointwise error at most ε satisfies*

$$\mathbb{E}\, g_{;}(C) \geq (1 - 2\varepsilon)2^{k-1}.$$

In particular, each exact circuit at each such size needs at least 2^{k-1} composition gates, even when chosen specifically for n.

The size threshold is $O(k4^k)$. Section 4.3 gives the analogous F_k bound and sharper operation-specific statements. The factor $1 - 2\varepsilon$ comes from comparing two inputs: a circuit that cannot distinguish them must err on at least one. It is attained for the one-sided projection and FO^2 quantifier counts of G_k.

Three versus two variables. The introductory expansions define both source queries in FO^2. For example, writing $H_A(x, y)$ for the conjunction defining H_A gives

$$G_k(x, y) \equiv \bigvee_{A \subseteq [k]} [H_A(x, y) \wedge \exists y\, H_{A^c}(x, y)].$$

The first conjunct checks the original read edge. In the second, that endpoint y is no longer needed, so its name can be reused for the witness; the binding does not extend to the first conjunct. The short FO^3 formula keeps the read endpoint and witness in scope together and can express their complementary requirements without this expansion. Section 5 uses the linear FO^2-to-CoR translation [Nak22] to show that every FO^2 formula defining G_k has at least 2^k quantifiers and $k2^k$ relation-atom occurrences. The expansion attains the quantifier bound exactly, and the minimum tree size is $\Theta(k2^k)$.

A lower bound for a binary query need not survive closing its free variables. Here the reference input satisfies G_k at every distinct pair, while each modified input has a failing pair. The tests therefore distinguish the sentences

$$G_k^{\forall} = \forall x\, \forall y\, (x = y \vee G_k(x, y)).$$

Theorem 1.4 (Three-variable versus two-variable sentences)**.** *For each $k \geq 1$, the sentence $G_k^{\forall}$ has size $O(k)$ and quantifier depth three over $2k$ binary relations. It is FO^2-definable, but every equivalent FO^2 sentence over finite structures has at least 2^k quantifiers and $k2^k$ relation-atom occurrences. Its minimum FO^2 tree size is $\Theta(k2^k)$.*

Substituting the displayed FO^2 expansion into $G_k^{\forall}$ uses $2^k + 2$ quantifiers, so the sentence quantifier bounds differ by at most two. On pure finite linear orders, three-variable sentences admit polynomial-size two-variable translations [GS05]; our exponential gap concerns general finite structures and growing binary signatures. Section 5.3 explains its relation to the variable-saving results of Vinall-Smeeth [VS24].

An elementary row-counting argument already gives a 2^{k-1} quantifier lower bound for the binary query G_k (Remark 5.4). The transport proof gives the exact binary-query quantifier count and the tree-size bound at every sufficiently large fixed domain size; its tests also survive passage to sentences and pointwise bounded error. There is also an independent exponential separation in the reverse direction: Proposition 5.3 gives a CoR term of size $O(k)$ that is FO^2-definable but needs exactly 2^k quantifiers in that logic. Since CoR terms translate linearly into FO^3, this elementary example also gives an FO^3-to-FO^2 gap for binary queries. Its existential closure has a constant-size FO^2 definition, illustrating why a binary-query separation alone does not establish a sentence separation.

Arity elimination in matrix query languages. Brijder, Gyssens, and Van den Bussche asked whether eliminating three-attribute intermediate relations in favor of matrix expressions necessarily incurs an exponential increase in size [BGV19, p. 9]. We prove such an increase over every nontrivial commutative idempotent semiring $(S, \vee, \otimes, 0_S, 1_S)$ of finite *join breadth* β: every finite join equals a join of at most β of its summands. Write $\mathcal{G}_k$ for G_k with disjunction, conjunction, and existential quantification replaced by $\vee$, $\otimes$, and the join over witnesses. It has a positive ARA(3) expression of size $O(k)$, where ARA(3) denotes annotated relational algebra with at most three attributes per intermediate relation.

Theorem 1.5 (Exponential arity-elimination overhead)**.** *For every such semiring and $k \geq 1$, any typed matrix circuit computing $\mathcal{G}_k$ on all finite inputs requires at least $2^k/(3\beta)$ matrix–vector, vector–matrix, or matrix–matrix product gates in total. This holds with arbitrary uniform entrywise functions and arbitrary two-input aggregation kernels in the model of Section 6.*

Circuits have fixed wiring and no vertex-dependent constant arrays; a kernel receives only its two scalar operands. Vectors, scalars, transpose, and diagonal operations are also allowed. For max–min and tropical semirings, $\beta = 1$. Finite join breadth supplies a bounded set of supporting colors for each aggregate, extending the local preservation argument beyond Boolean values. Section 1.3 explains this extension; Section 6 gives sharper operation counts, fixed-size and randomized bounds, and the precise model limits.

1.5 Related work

Relation algebra and its representations. The algebra of relations developed through work of De Morgan, Peirce, and Schröder; Pratt and Maddux survey its history [Pra92, Mad91]. Tarski's characterization identifies the binary queries expressible with three variable names [Tar41, TG87, Mad06]. For a palette of N colors, our reference color tables form the finite relation algebra $E_{N+1}(1, 2, 3)$. Jipsen, Maddux, and Tuza gave random and affine-plane representations of this algebra [JMT95]. Their affine-plane coloring already has our two-witness property when the plane has order at least four; Appendix A verifies this property. The probabilistic construction in Section 3.1 supplies the two-witness property separately at every sufficiently large domain size, which is needed for our size-specific bounds. We use these representations as reference inputs. The

additional ingredient for the lower bound is quantitative control of transport under hub recolorings, uniformly over all intermediate values; the existence of the finite representations alone does not provide an operation bound.

Succinctness and lower-bound methods. The classical FO^3-to-CoR translation has a single-exponential size bound, whereas the reverse translation from terms is linear; Appendix F gives the size analysis used here. In prior work, an exponential lower bound was established for positive CoR using two-edge witness queries and extended to the existential fragment Σ_1CoR [Nak22, Section 6]. That section also treats relational circuit size and gives a superpolynomial separation over a single binary relation symbol. Thus sharing was already covered for restricted target languages; the succinctness question for full CoR remained open. Our bounds allow arbitrary complement, converse, and shared intermediate results together. Theorem E.1 also gives the full-CoR lower bound for the source family of that work. Our main families use growing binary signatures, so the single-relation result and our main theorem address different restrictions. For F_k, the matching order of the composition count follows by implementing Hansel's monotone identification method as a fixed relational circuit [Han66, AS17].

Formula-size games and related potentials give separations between finite-variable, modal, and temporal logics [AI03, EVW02, HV19]. These methods already handle negation. Our proof instead exploits the finite algebra of reference values to bound the cost of distinct operations in a circuit, including circuits with unrestricted complement. For the variable-count comparison, Grohe and Schweikardt prove a polynomial translation from three-variable to two-variable sentences on pure finite linear orders [GS05, Theorem 4.3]. Vinall-Smeeth's variable-saving separations have at least three target variables and use higher-arity relations [VS24, Theorem 2]. Our sentence separation concerns three versus two variables on general finite structures with growing binary signatures. The linear FO^2-to-CoR translation is recorded in prior work [Nak22]; Proposition 5.3 shows an exponential cost in the other direction, even for terms that are FO^2-definable.

Matrices and query languages. Logical-to-algebraic translation is fundamental in databases [Cod70, Cod72], including graph query languages [FGL$^+$15, HPV$^+$21, HVGW22]. Relational composition is Boolean matrix multiplication, and fast matrix multiplication underlies algorithms for graph properties and first-order queries [Wil14, GIKW17, BFK19]. Our bounds count whole-relation operations in a query program; classical scalar-gate lower bounds for matrix products concern a different computation model [JS82, Juk12].

MATLANG combines matrix products, transpose, vector and diagonal operations, and entrywise functions [BGVW19, GMR$^+$21]. For semiring-annotated relations [GKT07], Brijder, Gyssens, and Van den Bussche connect binary-output three-attribute relational algebra with matrix expressions [BGV22]. Their earlier manuscript asks whether exponential arity-elimination overhead is necessary [BGV19, p. 9]. Our weighted query gives such a lower bound over each fixed nontrivial commutative idempotent semiring of finite join breadth, even with arbitrary aggregation kernels in the typed circuit model of Section 6. That section specifies the permitted operations and separately treats positive nonidempotent computations.

1.6 Organization

Section 2 fixes semantics and resource measures. Section 3 builds the reference graph, proves the two-color preservation lemma, and derives the test–cost theorem. Section 4 supplies the tests for G_k and F_k. Section 5 transfers the bound to quantifiers and adds the tree charging argument, the sentence result, and the independent reverse separation. Section 6 extends the local cost calculation to matrix languages. Section 7 collects the remaining questions. The appendices provide explicit graphs, the fixed Hansel circuit, tree and model-limit arguments, weighted F_k, directed and two-edge witness queries, and the classical single-exponential translation bound.

2 Preliminaries

Structures and formulas. A signature is a finite set of binary relation symbols, without function or individual constant symbols. A structure $\mathcal{M}$ has a nonempty domain $\mathrm{dom}(\mathcal{M})$ and interprets each symbol R by a binary relation $R^{\mathcal{M}}$. Lower bounds require correctness only on finite structures; the explicit translations used for upper bounds hold on all nonempty structures. The logics FO^2 and FO^3 use only the variable names x, y and x, y, z, respectively, allowing repeated binding of names. Equality is available. The free-variable set of φ is $\mathrm{FV}(\varphi)$. If it is contained in $\{x, y\}$, write $\varphi^{\mathcal{M}} \subseteq \mathrm{dom}(\mathcal{M})^2$ for its assignment relation. In particular, a sentence denotes the full or empty relation. All syntax uses unary negation and quantification and binary conjunction and disjunction; displayed iterated operations are abbreviations. The size $|\varphi|$ counts nodes after abbreviations are expanded. Thus $|F_k| = 4k$ and $|G_k| = 8k$. Each relation-atom occurrence has unit cost, regardless of the spelling of its relation name. This is a syntax-node measure over growing signatures, rather than a bit-length measure for encoded symbol names.

Evaluation and three-attribute intermediates. There is a direct $O(|\varphi|n^3)$ evaluation algorithm for an FO^3 formula on an n-element domain: store the truth table of each subformula over all assignments to x, y, z. Boolean operations act entrywise. For an existential quantifier, first take the disjunction along its coordinate and then repeat the resulting table along that coordinate. Each step takes $O(n^3)$ time, and universal quantification uses conjunction instead. In particular, G_k can be evaluated in $O(kn^3)$ time.

The same construction gives a linear-size translation into set relational algebra with at most three attributes per intermediate relation, using the unary domain relation as a constant. Lift every subformula to the common schema $\{x, y, z\}$. For a reflexive atom $R(u, u)$, select the diagonal of R and project to the attribute u; equality atoms are obtained by selection from domain products. Rename ordinary binary atoms as needed, and add missing domain coordinates by Cartesian product. Negation takes the difference from the full three-attribute domain relation. Existential quantification projects away its attribute and then restores that attribute by a Cartesian product with the domain. Each logical operation therefore uses a constant number of algebraic operations, and no intermediate has more than three attributes. For a binary assignment query, finally project to $\{x, y\}$, retaining either coordinate as a dummy if it is not free. This comparison concerns the permitted operations and schemas; our lower bounds below concern computation by the specified whole-relation or whole-matrix operations, rather than tuple-by-tuple evaluation time.

Terms and circuits. The CoR operations and constants have the meanings specified in the introduction. A *term* T is a syntax tree over binary input symbols, $\mathbf{0}, \mathbf{1}, \mathbf{id}$, union, intersection, complement, converse, and composition. Its value in $\mathcal{M}$ is $T^{\mathcal{M}}$. Its size $|T|$ is its number of nodes, and $\#_{;}(T)$ counts composition occurrences. A *relational circuit (DAG)* C uses the same operations and one designated output, but may share intermediate results. Write $|C|$ for its number of distinct nodes, including inputs, and $g_{;}(C)$ for its number of distinct composition gates. We subsequently say simply *relational circuit*. All operations other than composition may occur without any bound. For a tree,

$$|T| \geq 2\#_{;}(T) + 1,$$

since its leaves outnumber its binary nodes by one. For a term T, $\#_{\mathrm{rel}}(T)$ counts occurrences of input-relation symbols. For a logical formula φ, $\#_{\mathrm{rel}}(\varphi)$ counts occurrences of relation atoms, including reflexive atoms $R(x, x)$, and excludes equality atoms. Quantifier occurrences are counted by $\#_{\exists\forall}(\varphi)$; in a formula circuit we instead count distinct quantifier gates, denoted $g_{\exists\forall}(C)$. Its gates carry variable names and have the usual assignment semantics, so a subformula with given free variables can be shared without renaming.

Write $T \equiv_{\mathrm{fin}} \varphi$ when they define the same binary query on every finite structure over the input signature, and similarly for circuits and formulas. Unsubscripted equivalences used in explicit expansions hold on all nonempty structures. A size-specific circuit may depend on $n = |\operatorname{dom}(\mathcal{M})|$, but not on the input interpretations. No ordering, vertex predicates, named vertices, or arbitrary fixed $n \times n$ arrays are supplied for free. The only relational constants are the empty, full, and identity relations.

Randomized targets. A randomized circuit is an input-independent distribution of deterministic circuits over the permitted operations; its description may depend on n. Pointwise error at most ε means that, for every input and each output position, the reported value is correct with probability at least $1 - \varepsilon$. This need not be simultaneous correctness of the entire output relation. All randomized resource bounds below concern expectations over this distribution, whose support may contain incorrect or differently sized circuits. The same convention applies to randomized formulas and scalar computations.

Set and scalar notation. We use $[k] = \{1, \ldots, k\}$, $2^{[k]}$ for its power set, $\binom{[k]}{j}$ for its j-element subsets, and $B_k = \binom{k}{\lfloor k/2 \rfloor}$. For $I, A \subseteq [k]$, $I^c = [k] \setminus I$ and $A^c = [k] \setminus A$. Empty relation intersections and unions mean $\mathbf{1}$ and $\mathbf{0}$; empty conjunctions and disjunctions mean true and false. Relation complement is always denoted by an overline and is relative to the full square of the current domain. Truth values are $\mathbb{B} = \{0, 1\}$, and brackets $[\mathcal{P}]$ mean the indicator of a statement $\mathcal{P}$. In the construction, V is the vertex set, h its hub, Λ its palette, and $N = |\Lambda|$. General colors are λ, μ, ν; a target color identified with a subset of $[k]$ is written A. Transport is written T_f, to distinguish it from a term T. The letter d is reserved for the singleton selector bound in Section 5.2.

3 Recoloring and the test–cost theorem

We first prove the Boolean transport lemma. We then state the single counting theorem used throughout the paper. Its hypotheses separate the tests supplied by a query from the local

preservation properties of a target language. Section 6 will supply the same hypotheses for scalar-valued computations.

3.1 One finite colored graph

Fix a palette Λ of $N \geq 2$ colors and a vertex set V of size n. A coloring $\chi : \binom{V}{2} \to \Lambda$ assigns a color to every unordered pair of distinct vertices. We write $\chi(u, v)$ for $\chi(\{u, v\})$.

Lemma 3.1 (Two witnesses)**.** *For every integer*

$$n \geq n_0(N) := \left\lceil 12N^2 \ln(2N) \right\rceil + 3,$$

there is a coloring such that, for all distinct $u, v \in V$ and all $\lambda, \mu \in \Lambda$, at least two distinct vertices $z \in V \setminus \{u, v\}$ satisfy $\chi(u, z) = \lambda$ and $\chi(z, v) = \mu$.

Proof. Color the edges independently and uniformly. For fixed distinct u, v, w and colors λ, μ, the probability that no suitable witness occurs outside $\{u, v, w\}$ is at most $\exp(-(n-3)/N^2)$: the candidate witnesses use disjoint pairs of edges. A union bound gives failure probability at most

$$n^3 N^2 \exp(-(n-3)/N^2).$$

At $n = n_0(N)$, the inequality $n_0(N) \leq 13N^3$ bounds this expression by $2197/(4096N) < 1$. It decreases for $n > 3N^2$, proving the assertion at every larger size. A witness outside every additional forbidden vertex implies the required two-witness property. □

This is a two-witness formulation of the colored extension construction of Jipsen, Maddux, and Tuza [JMT95], who gave both random and affine-plane colorings. Their affine-plane construction already supplies two witnesses when the plane has order at least four. We state the property in this form because our proof must preserve profiles after excluding the hub. Proposition A.1 in Appendix A gives a self-contained two-witness affine-plane construction on fewer than $16N^2$ vertices, and a deterministic construction at every size $n \geq n_0(N)$. The quadratic scale is necessary: for fixed u, v, the N^2 profiles have disjoint witness sets of size at least two. In the countable homogeneous limit of finite edge-colored graphs [Hod93], every pair of distinct vertices has infinitely many witnesses for every pair of colors; Lemma 3.1 keeps only the finite fragment of this property that our proof uses.

Fix such a coloring, and then choose a vertex $h \in V$, called the hub. Every vertex sees every color. Moreover, every vertex other than h sees every color on edges avoiding h. No vertex, color, or color neighborhood is supplied as an additional input predicate.

3.2 Color tables and transport

For $\lambda \in \Lambda$, let $e_\lambda = \{(u, v) \in V^2 : u \neq v,\ \chi(u, v) = \lambda\}$. A *color table* $p = (p_{\mathbf{id}}, (p_\lambda)_{\lambda \in \Lambda})$ represents the Boolean relation X with value $p_{\mathbf{id}}$ on the diagonal and $p_{\chi(u,v)}$ at distinct u, v. Reference inputs will have this form; write $\mathcal{M}_0$ for the resulting structure.

Lemma 3.2 (Closure of color tables)**.** *The constants* $\mathbf{0}, \mathbf{1}, \mathbf{id}$ *have color tables, and relations with color tables are closed under Boolean operations, converse, and composition. If* X *and* Y *have tables* p *and* q*, then* $X;Y$ *has the table* r *given by*

$$\begin{aligned} r_{\mathbf{id}} &= (p_{\mathbf{id}} \wedge q_{\mathbf{id}}) \vee \bigvee_{\lambda\in\Lambda} (p_\lambda \wedge q_\lambda), \\ r_\lambda &= (p_{\mathbf{id}} \wedge q_\lambda) \vee (p_\lambda \wedge q_{\mathbf{id}}) \vee \left[\left(\bigvee_{\mu\in\Lambda} p_\mu \right) \wedge \left(\bigvee_{\nu\in\Lambda} q_\nu \right) \right]. \end{aligned} \tag{CT}$$

Proof. Boolean operations act entry by entry, and converse preserves color tables because χ is symmetric. For composition, endpoint witnesses give the first terms of (CT). By Lemma 3.1, every pair of distinct vertices has, for every pair of colors, a witness distinct from both endpoints; this supplies all color pairs off the diagonal. Likewise every vertex has a neighbor of every color, which supplies all equal-color pairs on the diagonal. □

In particular, every reference intermediate of a relational circuit has a color table, and

$$e_\lambda; e_\mu = \begin{cases} \mathbf{1}, & \lambda = \mu, \\ \overline{\mathbf{id}}, & \lambda \neq \mu. \end{cases}$$

Thus the color-table algebra is the symmetric relation algebra $E_{N+1}(1,2,3)$ of Jipsen, Maddux, and Tuza [JMT95]; here $\overline{\mathbf{id}} = \bigcup_{\lambda\in\Lambda} e_\lambda$ is the diversity relation. The proof needs the displayed multiplication table, not automorphisms of the chosen finite graph.

Given any map $f : \Lambda \to \Lambda$, recolor just the edges incident to h:

$$\chi_f(u,v) = \begin{cases} f(\chi(u,v)), & h \in \{u,v\}, \\ \chi(u,v), & h \notin \{u,v\}, \end{cases} \qquad (u \neq v).$$

For a reference value X with table p, define its *transport* $\mathsf{T}_f X$ by evaluating the same table on χ_f:

$$(\mathsf{T}_f X)(u,v) = \begin{cases} p_{\mathbf{id}}, & u = v, \\ p_{\chi_f(u,v)}, & u \neq v. \end{cases}$$

Transporting every input relation defines the recolored structure $\mathcal{M}_f$ on the same domain V. This does not assume closure of the independently reevaluated computation on $\mathcal{M}_f$.

3.3 The local calculation

For reference operands X, Y with tables p, q, and $J \subseteq \Lambda$, put

$$j_J^{\mathrm{L}} = \bigvee_{\lambda\in J} p_\lambda, \qquad j_J^{\mathrm{R}} = \bigvee_{\lambda\in J} q_\lambda, \qquad j_J^{\mathrm{D}} = \bigvee_{\lambda\in J} (p_\lambda \wedge q_\lambda).$$

Empty disjunctions have value zero. Let $J = f(\Lambda)$. At a pair of color λ, write $e(\lambda) = (p_{\mathbf{id}} \wedge q_\lambda) \vee (p_\lambda \wedge q_{\mathbf{id}})$ for the endpoint contribution. The following table gives the full product of the transported operands. Here u and v range over vertices other than h.

Output pair	$(\mathsf{T}_f X; \mathsf{T}_f Y)$ at that pair
$u \neq v$, $u, v \neq h$	$e(\chi(u,v)) \vee (j_\Lambda^{\mathrm{L}} \wedge j_\Lambda^{\mathrm{R}})$
(u,u), $u \neq h$	$(p_{\mathbf{id}} \wedge q_{\mathbf{id}}) \vee j_\Lambda^{\mathrm{D}}$
(h,v)	$e(\chi_f(h,v)) \vee (j_J^{\mathrm{L}} \wedge j_\Lambda^{\mathrm{R}})$
(u,h)	$e(\chi_f(u,h)) \vee (j_\Lambda^{\mathrm{L}} \wedge j_J^{\mathrm{R}})$
(h,h)	$(p_{\mathbf{id}} \wedge q_{\mathbf{id}}) \vee j_J^{\mathrm{D}}$

For distinct non-hub endpoints, every color pair is realized outside h, so the changed witness through h is redundant. For a non-hub diagonal pair, every equal-color pair is likewise realized outside h. At a hub row or column, the available profiles are exactly $J \times \Lambda$ or $\Lambda \times J$. At (h,h) they are the matching pairs indexed by J. This proves every entry of the table, including the diagonal cases.

Lemma 3.3 (Two colors per composition)**.** *For every pair of reference values X, Y with color tables p, q, there is a set $K = K(p,q) \subseteq \Lambda$, with $|K| \leq 2$, such that for every map $f : \Lambda \to \Lambda$,*

$$K \subseteq f(\Lambda) \quad \Longrightarrow \quad (\mathsf{T}_f X); (\mathsf{T}_f Y) = \mathsf{T}_f(X;Y).$$

If either operand is constant across the off-diagonal colors, K can be chosen with $|K| \leq 1$; if both are, $K = \varnothing$.

Proof. If some color has $p_\lambda = q_\lambda = 1$, retain that color. Otherwise retain a color with $p_\lambda = 1$, if one exists, and a color with $q_\lambda = 1$, if one exists. These choices preserve the three joins in the table. The image of f need only contain the chosen colors; f need not fix them.

For the sharper assertion suppose that all q_λ equal a bit. If that bit is zero, all generic contributions vanish. If it is one, only $\bigvee_{\lambda \in J} p_\lambda$ needs to be preserved, which requires at most one color. If both operands are color-independent, every nonempty image gives the same joins. The other orientation is symmetric. □

Uniform entrywise Boolean operations, converse, and the constants $\mathbf{0}, \mathbf{1}, \mathbf{id}$ commute with transport without any condition.

3.4 Tests and canonical cost

Choose distinct target colors in a set $\mathcal{H} \subseteq \Lambda$ and put $t = |\mathcal{H}|$. For each $a \in \mathcal{H}$, choose $\lambda_a \in \Lambda \setminus \{a\}$ and let f_a send a to λ_a and fix every other color. Write $\mathcal{M}_a = \mathcal{M}_{f_a}$ and $\mathsf{T}_a = \mathsf{T}_{f_a}$. We call this change a *one-color fold*.

A test for a query consists of $\mathcal{M}_0, \mathcal{M}_a$ and an output observation ρ_a whose two correct values differ, and for which

$$\rho_a(\mathsf{T}_a U) = \rho_a(U) \tag{O}$$

for every reference output value U. For binary queries, select $v_a \neq h$ with $\chi(h, v_a) \neq a$ and use $\rho_a(U) = U(h, v_a)$. The read edge is unchanged. Its color need not equal λ_a, although our applications choose them equal. Repeating observations for a single erased color does not increase t.

The counting argument uses only three properties of this construction. We call data satisfying them a *test system*:

(i) Each type of value has a collection of reference values closed under the permitted operations, and a transport map T_a for each test.

(ii) Every input on $\mathcal{M}_a$ equals the transport of its reference input on $\mathcal{M}_0$.

(iii) For each test, the output observation satisfies the observation condition (O) on every reference output value, and the required answers on $\mathcal{M}_0, \mathcal{M}_a$ differ.

For relational circuits, Lemma 3.2 supplies (i). The typed scalar computations in Section 6 will satisfy the same conditions. Transport maps carry the type of their argument; we suppress that type in the notation. For a gate g of a deterministic circuit C, let op_g be its operation and $\mathbf{U}_g$ the tuple of its operand values on $\mathcal{M}_0$. Transport acts componentwise on this tuple. Define

$$D_g = \{a \in \mathcal{H} : \mathrm{op}_g(\mathsf{T}_a \mathbf{U}_g) \neq \mathsf{T}_a(\mathrm{op}_g(\mathbf{U}_g))\},$$
$$c(g) = |D_g|, \qquad \mathrm{cost}(C) = \sum_{g \text{ distinct}} c(g).$$

For input nodes put $D_g = \varnothing$. Constant gates are included as nullary operations. The inequality defining D_g compares the entire output of that gate. Its left side uses transported *reference* operands, not independently reevaluated operands on $\mathcal{M}_a$.

This integer cost depends on the reference input and designated folds, which remain fixed, but is defined even for an incorrect circuit. A set K certifying commutation whenever $K \subseteq f(\Lambda)$, as in Lemma 3.3, gives the bound $c(g) \leq |K|$; it is not the definition of cost.

Theorem 3.4 (Test–cost theorem)**.** *Fix a test system with t tests. A deterministic circuit C correct at both observations of every test satisfies*

$$t \leq \mathrm{cost}(C).$$

If C_ω is chosen from an input-independent distribution of circuits, and each observation in each test is incorrect with probability at most $\varepsilon < 1/2$, then

$$(1 - 2\varepsilon)t \leq \mathbb{E}_\omega \, \mathrm{cost}(C_\omega).$$

Proof. Fix a deterministic circuit, not assumed correct, and put $K_C = \bigcup_g D_g$. For $a \notin K_C$, induction in a topological ordering of the gates gives

$$C(\mathcal{M}_a) = \mathsf{T}_a(C(\mathcal{M}_0)).$$

The equality concerns the whole output. By the observation condition (O), the two observations are equal, whereas their correct values differ. Hence at least one is wrong. If $\mathrm{err}_{a,0}(C), \mathrm{err}_{a,1}(C)$ are the corresponding zero-one error indicators, then

$$\mathrm{cost}(C) + \sum_{a \in \mathcal{H}} \big(\mathrm{err}_{a,0}(C) + \mathrm{err}_{a,1}(C)\big) \geq t,$$

because $|K_C| \leq \mathrm{cost}(C)$. Zero error proves the first assertion. Take expectations using the same random circuit on the two inputs of each test. Each error indicator has expectation at most ε, which proves the second assertion. No independence among the tests or their errors is required. □

This is the counting argument for all subsequent recoloring bounds. If $\varepsilon_{a,j}$ bounds the error at observation $j \in \{0,1\}$ of test a, the error-count inequality in the proof gives $\mathbb{E}\,\mathrm{cost}(C_\omega) \geq \sum_a (1-\varepsilon_{a,0}-\varepsilon_{a,1})$. In particular, let $\bar{\varepsilon}$ be the average error probability under the uniform distribution on the $2t$ input–observation pairs. Then $\mathbb{E}\,\mathrm{cost}(C_\omega) \geq t(1-2\bar{\varepsilon})$. For scalar outputs, an error means failure to return the exact required value. Pointwise correctness suffices; simultaneous correctness of all entries is a stronger assumption.

For ordinary CoR, write $S_X = \{\lambda \in \Lambda : p_\lambda = 1\}$ for the off-diagonal support of a reference operand X. Call a composition $X;Y$ *one-sided* if S_X or S_Y is $\varnothing$ or Λ. Let $m_1(C)$ count these gates and $m_2(C)$ the other composition gates, counting shared gates once.

Lemma 3.5 (Cost of compositions)**.** *For the fixed test system, every relational circuit C satisfies*

$$\mathrm{cost}(C) \leq m_1(C) + 2m_2(C) \leq 2g_;(C).$$

Proof. Lemma 3.3 bounds the cost of a one-sided composition by one and that of any other composition by two. Every other gate has cost zero. Summing over distinct gates proves the inequalities. □

The projections $X;\mathbf{1}$ and $\mathbf{1};X$ are one-sided. If a syntactic count is preferred, classify only these projections as one-sided and put every remaining composition in m_2; Lemma 3.5 still holds.

4 Two source queries

The construction of Section 3 is independent of the query. We now specify its input color tables. In each case the read edge identifies the witness color which a fold removes.

4.1 Complementary profiles

For G_k, take $\Lambda = \mathcal{H} = 2^{[k]}$. On a color μ, interpret P_i as 1 precisely when $i \in \mu$, and Q_i as 1 precisely when $i \notin \mu$. Put both values zero on every loop. More explicitly,

$$P_i^{\mathcal{M}_0} = \bigcup_{\mu\in\Lambda:\, i\in\mu} e_\mu, \qquad Q_i^{\mathcal{M}_0} = \bigcup_{\mu\in\Lambda:\, i\notin\mu} e_\mu.$$

For each target $A \in \mathcal{H}$, choose v_A with $\chi(h, v_A) = A^c$, and use the fold

$$\lambda_A = A^c, \qquad f_A(A) = A^c, \qquad f_A(\mu) = \mu \quad (\mu \in \Lambda \setminus \{A\}).$$

Here $A^c = [k] \setminus A$ and $A \neq A^c$ for $k \geq 1$. The read edge is therefore unchanged.

Lemma 4.1 (The tests for G_k)**.** *For every $A \subseteq [k]$,*

$$G_k^{\mathcal{M}_0}(h, v_A) = 1, \qquad G_k^{\mathcal{M}_A}(h, v_A) = 0.$$

Thus G_k supplies $t = 2^k$ tests.

Proof. For non-loop edges with colors μ, ν, the body of G_k holds exactly when $\nu = \mu^c$. A loop, whose P_i, Q_i values are all zero, cannot be a witness edge. Against the read color A^c, the sole accepted witness color is A. That color occurs at h in $\mathcal{M}_0$ and does not occur there after the fold. □

4.2 Disjunctive clauses

For F_k, where $k \geq 2$, take

$$\Lambda = \binom{[k]}{\lfloor k/2 \rfloor} \cup \binom{[k]}{\lceil k/2 \rceil}, \qquad \mathcal{H} = \binom{[k]}{\lceil k/2 \rceil}, \qquad R_i^{\mathcal{M}_0} = \bigcup_{\mu \in \Lambda:\, i \in \mu} e_\mu.$$

All input loops again have value zero. Use the same read colors and folds as in Section 4.1. If k is even, the palette is one layer. If k is odd, it has two layers, but only the upper layer supplies tests.

Lemma 4.2 (The tests for F_k)**.** *For every $A \in \mathcal{H}$,*

$$F_k^{\mathcal{M}_0}(h, v_A) = 1, \qquad F_k^{\mathcal{M}_A}(h, v_A) = 0.$$

Thus F_k supplies $t = B_k = \binom{k}{\lfloor k/2 \rfloor}$ tests.

Proof. A witness edge of color $\mu \in \Lambda$ is accepted precisely when $A^c \cup \mu = [k]$, or $A \subseteq \mu$. All palette sets have size at most $|A|$, so this holds precisely for $\mu = A$. Loop witnesses fail because A is nonempty. The fold removes exactly this accepted color from the hub row. □

Here a *singleton selector* is a Boolean formula in the input coordinate profiles and constants whose support on the entire palette is $\{a\}$ for some target color $a \in \mathcal{H}$; d is a lower bound on its number of input-relation occurrences. The values in the following table are proved in Section 5.2. This parameter is needed only for the tree-size refinement.

Query	Palette size N	Tests t	Selector size d
G_k	2^k	2^k	k
F_k, even k	B_k	B_k	$k/2$
F_k, odd k	$2B_k$	B_k	$(k+1)/2$

4.3 Consequences and upper bounds

Theorem 1.1 (Exponential lower bounds for full CoR)**.** *Every relational circuit C and* CoR *term T equivalent on finite structures to a query below satisfy the indicated lower bounds:*

Query	*Composition gates $g_{\cdot}(C)$*	*Term size $\lvert T \rvert$*
F_k $(k \geq 2)$	$\lceil B_k/2 \rceil$	$2\lceil k/2 \rceil B_k - 1$
G_k $(k \geq 1)$	2^{k-1}	$2k2^k - 1$

The composition bounds also apply to terms.

Proof. Lemmas 4.2 and 4.1 supply B_k and 2^k tests. Theorem 3.4 and Lemma 3.5 therefore give the composition lower bounds, including for circuits with arbitrary sharing. Section 5.2 supplies the stated tree-size lower bounds. □

Together with the expansion of G_k in Section 1.1, these lower bounds show that the minimum circuit size for G_k is $\Theta(2^k)$ and its minimum term size is $\Theta(k2^k)$.

Corollary 1.3 (Fixed sizes and bounded error)**.** *Fix $k \geq 1$, $n \geq n_0(2^k)$, and $0 \leq \varepsilon < 1/2$. Every input-independent distribution of relational circuits computing G_k on all n-element inputs with pointwise error at most ε satisfies*

$$\mathbb{E}\, g_{;}(C) \geq (1 - 2\varepsilon)2^{k-1}.$$

In particular, each exact circuit at each such size needs at least 2^{k-1} composition gates, even when chosen specifically for n.

Proof. At the prescribed size n, Lemma 3.1 gives the reference graph, and Lemma 4.1 gives all 2^k tests on that domain. Apply the expected-cost assertion of Theorem 3.4 and $\mathrm{cost}(C) \leq 2g_{;}(C)$. Setting $\varepsilon = 0$ includes every exact size-specific circuit. □

More precisely, the same argument gives $\mathbb{E}(m_1 + 2m_2) \geq (1 - 2\varepsilon)t$ for both queries, with $(N, t) = (2^k, 2^k)$ for G_k and the values in the table for F_k. Each bound holds separately at every $n \geq n_0(N)$.

The expansions in the introduction give at most $2^k - 1$ compositions for F_k and 2^k for G_k. Shared prefix intersections implement the latter with $O(2^k)$ total nodes. For F_k, learning a monotone function for each row improves the composition count.

Proposition 4.3 (A fixed Hansel circuit)**.** *On all nonempty input structures, F_k has a converse-free relational circuit C with*

$$g_{;}(C) \leq B_{k+1}, \qquad |C| = O(2^k).$$

Every composition is a row projection. At most $\lfloor k/2 \rfloor + 1$ composition gates occur on any path.

The monotone identification bound is due to Hansel [Han66]; see Aslanyan and Sahakyan [AS17, Theorem 1] for the chain formulation. Appendix B gives a self-contained fixed-circuit implementation, including its Boolean control gates. Thus

$$\lceil B_k/2 \rceil \leq \min_{C \equiv_{\mathrm{fin}} F_k} g_{;}(C) \leq B_{k+1} \leq 2B_k.$$

The order of the composition count is determined. The corresponding total-size interval remains $\Omega(B_k)$ to $O(2^k)$. The corresponding construction for FO^2 formula circuits has at most B_{k+1} distinct quantifier gates (Appendix B).

Corollary 4.4 (Without converse)**.** *Every converse-free relational circuit defining F_k on finite structures has at least B_k composition gates, for $k \geq 2$. Every such circuit defining G_k has at least 2^k composition gates, for $k \geq 1$. The latter bound is exact.*

Appendix E.1 proves the corollary with a directed reference coloring. For full CoR, Proposition C.1 gives a circuit with 2^{k-1} compositions on the symmetric recoloring family itself. Thus that family attains the coefficient two in the local bound. The optimum for G_k on arbitrary inputs remains between 2^{k-1} and 2^k. Appendix E.2 also treats the two-edge witness queries from earlier succinctness work; the full-CoR lower bounds are stated in Theorem E.1.

For Boolean G_k, the factor $1 - 2\varepsilon$ is sharp for canonical cost. Its DNF has cost at most 2^k, and the theorem gives the reverse inequality. Run that exact circuit with probability $1 - 2\varepsilon$, and output **0** or **1** with probability ε each. Every entry then has error exactly ε, with expected cost

$(1-2\varepsilon)2^k$. All its compositions are projections, so $m_2 = 0$ and its expected one-sided composition count is also $(1-2\varepsilon)2^k$. The same construction attains the expected quantifier bound in Section 5, and the one-sided operation bounds for Boolean CoR and MATLANG. These equalities use the pointwise-error and expected-cost conventions of Section 2.

5 Two-variable logic and tree size

The formulas F_k, G_k are FO^2-definable, but their displayed linear-size forms use three variable names. This section quantifies the difference. We also give an elementary succinctness separation in the reverse direction, from CoR to FO^2.

5.1 A translation that preserves the relevant resources

For a formula φ using only x, y, view its value as a binary relation of assignments, even if one or both variables are absent from $\mathrm{FV}(\varphi)$. Thus a sentence denotes $\mathbf{1}$ or $\mathbf{0}$. The standard linear translation τ into CoR is as follows:

φ	$\tau(\varphi)$	φ	$\tau(\varphi)$
$R(x,y)$	R	$R(y,x)$	$R^\smile$
$R(x,x)$	$(R \cap \mathbf{id}); \mathbf{1}$	$R(y,y)$	$\mathbf{1}; (R \cap \mathbf{id})$
$x = y$ or $y = x$	$\mathbf{id}$	$x = x$ or $y = y$	$\mathbf{1}$
$\exists x\, \psi$	$\mathbf{1}; \tau(\psi)$	$\exists y\, \psi$	$\tau(\psi); \mathbf{1}$

Conjunction, disjunction, and negation translate to intersection, union, and complement. Universal quantifiers use $\forall u\, \psi \equiv \neg\exists u\, \neg\psi$, for $u \in \{x, y\}$. Induction on formulas proves correctness on all nonempty structures. The linear size bound is recorded in [Nak22, Remark 3.22].

Translation preserves the relation-atom count $\#_{\mathrm{rel}}$ exactly. For the reference structures of Sections 4.1–4.2, the off-diagonal support of $R \cap \mathbf{id}$ is empty. Hence the extra composition used for a reflexive atom has canonical cost zero. Every quantifier contributes a one-sided composition and has cost at most one. These observations give the following resource bound.

Lemma 5.1 (Quantifier cost)**.** *For either source-query test system, every* FO^2 *formula* φ *satisfies*

$$\mathrm{cost}(\tau(\varphi)) \le \#_{\exists\forall}(\varphi).$$

For a formula circuit, canonical cost is at most its number of distinct quantifier gates. The translation preserves relation-atom occurrences exactly for formulas.

This is a statement about canonical cost, not syntactic composition count: the latter also counts the zero-cost compositions introduced by reflexive atoms.

Combining Lemma 5.1 with Theorem 3.4 gives

$$\varphi \equiv_{\mathrm{fin}} G_k \Longrightarrow \#_{\exists\forall}(\varphi) \ge 2^k, \qquad \varphi \equiv_{\mathrm{fin}} F_k \Longrightarrow \#_{\exists\forall}(\varphi) \ge B_k.$$

The FO^2 expansions obtained from the introductory DNFs use exactly 2^k quantifiers for G_k and at most $2^k - 1$ for F_k. In particular, the optimum quantifier count for G_k is exactly 2^k. At a fixed size $n \ge n_0(N)$ these conclusions still hold. For a distribution of FO^2 formulas or circuits with pointwise error at most ε, the corresponding expected quantifier count is at least $(1-2\varepsilon)t$. No restriction to existential target formulas is made.

5.2 Counting input occurrences in trees

The circuit theorem also yields a stronger bound for syntax trees. A singleton selector for a target color $a \in \mathcal{H}$ is a Boolean formula in the input color profiles, with constants allowed, whose support on the entire palette is exactly $\{a\}$.

Lemma 5.2 (Disjoint input-occurrence charging)**.** *Fix the reference structure and one-color folds of Section 3.4. Suppose every singleton selector of a target color requires at least $d > 0$ input-relation occurrences. Then every* CoR *term T, whether correct or not, satisfies*

$$\mathrm{cost}(T) \le \#_{\mathrm{rel}}(T)/d.$$

Appendix C.1 proves the lemma by assigning disjoint sets of input occurrences to composition nodes. The disjointness uses the tree structure of a term.

For the cube used by G_k, omitting coordinate i leaves a color indistinguishable from its i-th-bit flip, even when both P_i, Q_i are available. Hence $d = k$. For F_{2m}, reading fewer than m coordinates leaves an unread element in each of A, A^c; swapping them preserves the observed profile. For F_{2m+1}, reading fewer than $m+1$ coordinates leaves an unread element of the target A; deleting it gives another palette color with the same observed profile. Thus $d = \lceil k/2 \rceil$ for F_k. The conjunction of the positive coordinates of A selects A in the F_k palette, and H_A selects A in the G_k palette, so these selector bounds are exact. Theorem 3.4, Lemma 5.2, and preservation of $\#_{\mathrm{rel}}$ under τ give

	F_k	G_k
$\#_{\mathrm{rel}}(T),\ \#_{\mathrm{rel}}(\varphi)$	$\ge \lceil k/2 \rceil B_k$	$\ge k2^k$

as lower bounds whenever T or $\varphi \in \mathrm{FO}^2$ defines the indicated query. A unary/binary syntax tree with L leaves has at least $2L-1$ nodes; each quantifier is an additional unary node. Therefore

$$\begin{aligned}
&|T| \ge 2\lceil k/2 \rceil B_k - 1 && (T \equiv_{\mathrm{fin}} F_k),\\
&|T| \ge 2k2^k - 1 && (T \equiv_{\mathrm{fin}} G_k),\\
&|\varphi| \ge (2\lceil k/2 \rceil + 1)B_k - 1 && (\varphi \in \mathrm{FO}^2,\ \varphi \equiv_{\mathrm{fin}} F_k),\\
&|\varphi| \ge (2k+1)2^k - 1 && (\varphi \in \mathrm{FO}^2,\ \varphi \equiv_{\mathrm{fin}} G_k).
\end{aligned}$$

Factoring the read-edge tests gives CoR and FO^2 trees for G_k of sizes at most $(2k+6)2^k - 5$ and $(2k+5)2^k - 5$, respectively (Appendix C.2). Thus both optimal tree sizes are $\Theta(k2^k)$. For F_k the established tree-size interval is $\Omega(kB_k)$ to $O(k2^k)$. Combining Lemma 5.2 with the randomized test–cost bound yields $\mathbb{E}\#_{\mathrm{rel}}(T) \ge d(1-2\varepsilon)t$ for randomized terms, and the same bound holds for randomized FO^2 formulas.

5.3 Sentences and the variable hierarchy

Recall the sentence $G_k^{\forall} = \forall x \forall y\,(x = y \vee G_k(x,y))$ from the introduction. We now prove the sentence separation stated in the introduction.

Theorem 1.4 (Three-variable versus two-variable sentences)**.** *For each $k \ge 1$, the sentence $G_k^{\forall}$ has size $O(k)$ and quantifier depth three over $2k$ binary relations. It is* FO^2*-definable, but every equivalent* FO^2 *sentence over finite structures has at least 2^k quantifiers and $k2^k$ relation-atom occurrences. Its minimum* FO^2 *tree size is $\Theta(k2^k)$.*

Proof. The source has size $8k + 4$ and quantifier depth three. It is true in $\mathcal{M}_0$: every off-diagonal read color has a complementary witness color, and equality handles the diagonal. It is false in every $\mathcal{M}_A$, as witnessed by (h, v_A).

A sentence is represented by a constant binary relation, and transport fixes either constant. Thus these structures give 2^k tests for the sentence itself. For any equivalent FO^2 sentence φ, apply Theorem 3.4 to $\tau(\varphi)$. Lemma 5.1 gives $\#_{\exists\forall}(\varphi) \geq 2^k$, and Lemma 5.2, together with exact preservation of $\#_{\mathrm{rel}}$, gives $\#_{\mathrm{rel}}(\varphi) \geq k2^k$. Conversely, substitute the FO^2 expansion of G_k into the sentence. It has $2^k + 2$ quantifiers and $O(k2^k)$ nodes. This proves FO^2-definability and the matching order of tree size. □

The quantifier optimum for this sentence is therefore in $[2^k, 2^k + 2]$; the exact 2^k optimum proved earlier concerns the binary query. The lower bounds hold separately at every $n \geq n_0(2^k)$. For randomized FO^2 sentences with error at most ε on each input, the same tests give expected quantifier count at least $(1 - 2\varepsilon)2^k$ and expected relation-atom count at least $(1 - 2\varepsilon)k2^k$.

This complements the variable-saving separations of Vinall-Smeeth [VS24, Theorem 2], whose stated range starts at three variables on the target side. Our result concerns the remaining comparison of three and two variables, over growing binary signatures. It does not claim a lower bound for separating a single prescribed pair of structures as a function of their size. On pure finite linear orders, in contrast, Grohe and Schweikardt [GS05, Theorem 4.3] give a polynomial translation from three-variable sentences to two-variable sentences. The qualifications "pure orders" and "sentences" are essential to that comparison.

5.4 An elementary separation from CoR to FO^2

The linear translation above goes from FO^2 to CoR. Even a FO^2-definable CoR term may be exponentially more succinct than FO^2. The following independent example requires no recoloring lemma.

Both this example and the remark below use a simple row-counting principle. Fix one structure and disjoint sets U, W of row and column vertices such that every atomic formula at $(u, v) \in U \times W$ has a value independent of u. Cut an FO^2 formula at the first quantifier on each path from its root. Every cut subformula has at most one free variable. If r of these occurrences have free-variable set exactly $\{x\}$, their r truth values determine the entire output row: the uncut atoms, sentences, and y-only subformulas are independent of the row vertex. Thus the formula has at most 2^r distinct output rows on $U \times W$. In particular, 2^s distinct rows require at least s quantifiers. The same argument counts distinct cut gates in a formula circuit.

Proposition 5.3 (A reverse succinctness gap)**.** *For every $k \geq 1$, over $k + 1$ binary input symbols there is a converse-free* CoR *term T_k of size $O(k)$ that is FO^2-definable, but every equivalent FO^2 formula over finite structures contains at least 2^k quantifiers. This quantifier bound is attained.*

Proof. Use input symbols $R, P_1, \ldots, P_k$. Put

$$D_i^1 = \mathbf{id} \cap P_i, \qquad D_i^0 = \mathbf{id} \cap \overline{P_i}, \qquad T_k = R; \bigcap_{i=1}^{k} \left[(D_i^0; \mathbf{1}; D_i^0) \cup (D_i^1; \mathbf{1}; D_i^1) \right].$$

The complement in D_i^0 is taken before intersection with the identity; it is not the complement of D_i^1. Expanding these abbreviations gives an $O(k)$-size tree. Its semantics is

$$\varphi_k(x,y) = \exists z \left(R(x,z) \wedge \bigwedge_{i=1}^{k} (P_i(z,z) \leftrightarrow P_i(y,y)) \right).$$

Thus $T_k(x,y)$ holds exactly when $\varphi_k(x,y)$ does. Here $\alpha \leftrightarrow \gamma$ abbreviates $(\alpha \wedge \gamma) \vee (\neg\alpha \wedge \neg\gamma)$. For $A \subseteq [k]$ and a variable name u, define

$$\vartheta_A(u) = \bigwedge_{i \in A} P_i(u,u) \ \wedge \bigwedge_{i \in A^c} \neg P_i(u,u).$$

The formula

$$\bigvee_{A \subseteq [k]} [\vartheta_A(y) \wedge \exists y \, (R(x,y) \wedge \vartheta_A(y))]$$

is an equivalent FO^2 formula with exactly 2^k quantifiers. The inner y is rebound only in the second conjunct.

For the lower bound put $\Lambda = 2^{[k]}$. Construct one finite structure with distinct vertices u_J for every $J \subseteq \Lambda$, and distinct vertices v_A, w_A for every $A \in \Lambda$; the three sets of vertices are disjoint. Interpret

$$R = \{(u_J, w_A) : A \in J\}, \qquad P_i = \{(v_A, v_A), (w_A, w_A) : i \in A\}.$$

All other input entries are false. Then $T_k(u_J, v_A) = 1$ exactly when $A \in J$. Thus the restricted output matrix has 2^{2^k} distinct rows.

At (u_J, v_A), every atomic formula is independent of J: mixed-variable equality is false and reflexive equalities are true, all direct R-edges between these vertices are absent, and all P_i-loops at u_J are false. The row-counting principle therefore gives at least 2^k quantifiers. □

The lower-bound structure has $2^{2^k} + 2^{k+1}$ vertices. The formula φ_k also gives an elementary exponential separation between FO^3 and FO^2 for binary queries: it has size $O(k)$ and one quantifier, whereas its minimum FO^2 quantifier count is 2^k. More generally, every CoR term translates linearly to a FO^3 formula by translating composition as existential quantification, reusing the third variable at each composition.

This binary-query example also illustrates why sentence lower bounds need a separate argument. Its existential closure collapses:

$$\exists x \, \exists y \, \varphi_k(x,y) \quad \Longleftrightarrow \quad \exists x \, \exists y \, R(x,y).$$

Indeed, any edge $R(x,z)$ satisfies the profile condition after choosing $y = z$. In contrast, the universal closure $G_k^{\forall}$ preserves all 2^k tests, as established in the proof of Theorem 1.4.

Remark 5.4 (An elementary bound for G_k). The same principle gives an exponential lower bound for G_k itself. For $k \geq 1$, put $\mathcal{A} = \{A \subseteq [k] : 1 \in A\}$ and $s = |\mathcal{A}| = 2^{k-1}$. Use disjoint vertices u_J for $J \subseteq \mathcal{A}$, and v_A, w_{A^c} for $A \in \mathcal{A}$. An edge of profile $\mu \subseteq [k]$ has values $P_i = [i \in \mu]$ and $Q_i = [i \notin \mu]$. Give (u_J, v_A) profile A for every J, A, and give (u_J, w_{A^c}) profile A^c precisely when $A \in J$. All other input entries are false. Since $\mathcal{A}$ contains no complementary pair, no read edge can serve as a witness for another read edge. Hence $G_k(u_J, v_A) = [A \in J]$. Atomic values at these read edges are independent of J, so the 2^s distinct output rows imply at least $s = 2^{k-1}$ quantifiers.

This structure has $2^{2^{k-1}} + 2^k$ vertices; adding isolated vertices preserves the argument at every larger size. The recoloring proof gives the exact quantifier bound 2^k already at every $n \geq n_0(2^k) = O(k4^k)$. It also gives the $k2^k$ relation-atom bound, the sentence separation, and the sharp expected-count factor $1 - 2\varepsilon$. These are stronger conclusions than the elementary row count alone establishes.

6 Scalar aggregation and matrix query languages

Relational composition aggregates two edge values over a common index. The preservation calculation extends to scalar aggregation without any order or monotonicity assumption on the operation combining those values. This gives arity-elimination lower bounds for matrix query languages.

6.1 Finite join breadth and arbitrary kernels

Let $(S, \vee, 0_S)$ be a join-semilattice with least element 0_S. Its *join breadth* β is the least nonnegative integer such that every finite join equals the join of at most β of its summands. The empty join is 0_S. We assume $\beta < \infty$. A totally ordered set with maximum as join has breadth one. The space $\{0,1\}^b$ with coordinatewise disjunction has breadth b.

Let V be the common finite index set of the input matrices, with $n = |V|$. For any function $\kappa : S \times S \to S$, called a *kernel*, define

$$(X ;_\kappa Y)(u,v) = \bigvee_{z \in V} \kappa(X(u,z), Y(z,v)).$$

Kernels may differ between gates. They need not be associative, monotone, or zero-preserving. We also allow arbitrary uniform entrywise functions, transpose, uniform scalar constants, and a diagonal mask with two fixed values, one on and one off the diagonal. "Uniform" means the same function is used at all positions of a given type; it has no access to vertex names. The inputs to the query are square $n \times n$ arrays. No order, vertex identifiers, arbitrary input vectors, or vertex-dependent constant arrays are supplied in addition to them.

Each circuit has a fixed directed acyclic wiring. Every entrywise function, kernel, and scalar constant is chosen before seeing the input; it may depend on n and k in a size-specific circuit. An entrywise function receives the scalar operands at its position, and a kernel receives only the two scalar arguments displayed above. In particular, a kernel does not also receive $X(u,v)$, the position names u, v, z, or an encoding of the whole input unless such a value has been computed and supplied through its two operands by permitted operations. Data-dependent values and tests are allowed, but there is no primitive for data-dependent rewiring or indexed access to arbitrary entries. These conventions specify the target model used in Theorem 1.5.

Reference arrays are represented by scalar color tables $p = (p_{\mathbf{id}}, (p_\lambda)_{\lambda \in \Lambda})$ just as in Section 3.2. Entrywise functions and transpose preserve them. For tables p, q and $J \subseteq \Lambda$, define

$$\kappa_J^{\mathrm{L}} = \bigvee_{\lambda \in J,\, \mu \in \Lambda} \kappa(p_\lambda, q_\mu), \qquad \kappa_J^{\mathrm{R}} = \bigvee_{\lambda \in J,\, \mu \in \Lambda} \kappa(p_\mu, q_\lambda),$$

$$\kappa_J^{\mathrm{D}} = \bigvee_{\lambda \in J} \kappa(p_\lambda, q_\lambda), \qquad \kappa_* = \kappa_\Lambda^{\mathrm{L}} = \kappa_\Lambda^{\mathrm{R}}.$$

The reference product has table

$$r_{\mathbf{id}} = \kappa(p_{\mathbf{id}}, q_{\mathbf{id}}) \vee \kappa^{\mathrm{D}}_{\Lambda},$$
$$r_{\lambda} = \kappa(p_{\mathbf{id}}, q_{\lambda}) \vee \kappa(p_{\lambda}, q_{\mathbf{id}}) \vee \kappa_{*}.$$

Transport evaluates the same table on the recolored graph. The five cases of Section 3.3 remain valid, with endpoint contributions formed by κ, full off-diagonal contribution κ_*, and the three possibly altered contributions $\kappa^{\mathrm{L}}_J, \kappa^{\mathrm{R}}_J, \kappa^{\mathrm{D}}_J$ at the hub row, column, and diagonal, respectively. This follows from the same witness profiles; idempotence of join makes multiplicities irrelevant.

Lemma 6.1 (Scalar local bounds)**.** *A scalar aggregation gate has canonical cost at most 3β. If one operand is constant across off-diagonal colors, its cost is at most β; if both are, its cost is zero. For Boolean scalars with arbitrary kernels, the general bound improves to two. For b-bit scalars with coordinatewise disjunction and arbitrary kernels it improves to $2b$.*

Proof. Choose at most β supporting summands for each of $\kappa^{\mathrm{L}}_{\Lambda}$, $\kappa^{\mathrm{R}}_{\Lambda}$, and $\kappa^{\mathrm{D}}_{\Lambda}$. From these summands retain, respectively, the row colors, the column colors, and the common colors on the diagonal. For example, a supporting summand of the first join chooses a pair (λ, μ), and only its first color λ must remain in J. Retaining their union preserves the product everywhere. Its size is at most 3β, and a fold can fail only if it deletes a color in this union.

If q_λ is independent of λ, then $\kappa^{\mathrm{L}}_J = \kappa^{\mathrm{D}}_J$ and $\kappa^{\mathrm{R}}_J = \kappa_*$ for every nonempty J. Only one join needs support. If both profiles are constant, all three joins are unchanged. For Boolean scalars, if $\kappa^{\mathrm{D}}_{\Lambda} = 1$, a single color attaining that value simultaneously supports both other joins. Otherwise the diagonal join is always zero, and at most one supporting color for each of the other joins suffices. Apply this argument independently to each output bit for the b-bit bound. □

These calculations provide reference closure and local cost bounds for Theorem 3.4; Section 6.3 supplies the query tests. Uniform entrywise functions, transpose, and the permitted constants have zero cost.

Remark 6.2 (Why three colors can be needed). The scalar coefficient three is attained locally even at breadth one. Use max–min on $[0, 2]$, and choose three distinct colors a, b, c with

$$(p_a, q_a) = (2, 0), \qquad (p_b, q_b) = (0, 2), \qquad (p_c, q_c) = (1, 1).$$

Set the other color values and both diagonal values to zero. A fold removing a changes the hub-row value from 2 to 1; removing b changes the hub-column value from 2 to 1; removing c changes the hub-diagonal value from 1 to 0. Thus one product has canonical cost three on these tests. The three affected positions explain the scalar bound. Proposition C.2 gives a matching asymptotic product count for the full symmetric recoloring promise, and Appendix C.6 also gives the corresponding b-bit packing.

6.2 Typed matrix operations

We use the dimension symbols n and 1. Intermediate arrays have shapes $n \times n$, $n \times 1$, $1 \times n$, or 1×1; these are matrices, column vectors, row vectors, and scalars, respectively. An admissible contraction of an $a \times n$ and an $n \times c$ array is

$$Z(i, j) = \bigvee_{z=1}^{n} \kappa(X(i, z), Y(z, j)), \qquad a, c \in \{1, n\}.$$

Products contracting an index of size one are just applications of the kernel and are not data-dimension contractions. We allow entrywise functions, transpose, scalar and vector broadcasting, diagonal extraction and formation, uniform constant vectors, and the diagonal mask. All of these are uniform and do not aggregate a data index. This is at least as expressive as the corresponding MATLANG fragment with semiring multiplication as kernel; standard MATLANG operations are described in [BGVW19, BGV22].

At the reference input, every vector is constant over its data index and every scalar is a single value. Encode a column vector $\mathbf{c}$ as the square array $(u, v) \mapsto \mathbf{c}(u)$, a row vector $\mathbf{r}$ as $(u, v) \mapsto \mathbf{r}(v)$, and a scalar as a constant square array. Reference closure follows from the color-table calculation. The encoded reference vector has the same value on every color and the diagonal, so transport acts identically on it. Diagonal formation and extraction and all broadcasting operations commute with transport. Matrix–vector and vector–matrix contractions need at most β colors by Lemma 6.1. A vector inner product has cost zero since both reference operands are constant and transport leaves them fixed. Products over a size-one index also have cost zero.

Let $\#_{\mathrm{mv}}(C)$ count matrix–vector and vector–matrix contractions, and let $\#_{\mathrm{mm}}(C)$ count square matrix–matrix contractions, always counting shared gates once. The local bounds give the following typed resource estimate.

Lemma 6.3 (Cost of typed products)**.** *Every typed matrix circuit C in the model above satisfies*

$$\mathrm{cost}(C) \le \beta(\#_{\mathrm{mv}}(C) + 3\#_{\mathrm{mm}}(C)).$$

For Boolean scalars and arbitrary kernels, the bound improves to $\#_{\mathrm{mv}}(C) + 2\#_{\mathrm{mm}}(C)$; for b-bit scalars it is $b(\#_{\mathrm{mv}}(C) + 2\#_{\mathrm{mm}}(C))$.

Proof. Sum the bounds of Lemma 6.1 over the distinct product gates. The other permitted operations commute with transport, as established above, and therefore have cost zero. □

The gate-by-gate accounting can be summarized as follows.

Operation	Canonical cost at most
Uniform entrywise function, transpose, permitted constant	0
Diagonal formation or extraction, broadcasting	0
Product over an index of size one	0
Vector inner product	0
Matrix–vector or vector–matrix contraction	β
Square matrix–matrix contraction	3β

Canonical cost measures failure to commute with the fixed transports; zero cost does not mean zero evaluation time. Inner products contribute zero to canonical cost.

6.3 Arity elimination for a weighted query

Let $(S, \vee, \otimes, 0_S, 1_S)$ be a nontrivial commutative idempotent semiring of finite join breadth β. Thus addition $\vee$ is an idempotent commutative monoid, multiplication $\otimes$ is a commutative monoid,

multiplication distributes over addition, and 0_S annihilates multiplication. For the common input index set V, define

$$\mathcal{G}_k(x,y) = \bigvee_{z\in V} \bigotimes_{i=1}^{k} \left[(P_i(x,y)\otimes Q_i(x,z)) \vee (Q_i(x,y)\otimes P_i(x,z))\right].$$

An annotated relation assigns a scalar in S to each tuple. In positive annotated relational algebra, union is pointwise addition, natural join multiplies annotations, and projection sums over removed attributes. Renaming changes attribute names. We write ARA(3) for this language restricted to intermediate relations with at most three attributes. The weighted query $\mathcal{G}_k$ has an ARA(3) expression of size $O(k)$: form the two ternary joins in each clause, take their union, join the clauses, and project out z.

Theorem 1.5 (Exponential arity-elimination overhead)**.** *For every such semiring and $k \geq 1$, any typed matrix circuit computing $\mathcal{G}_k$ on all finite inputs requires at least $2^k/(3\beta)$ matrix–vector, vector–matrix, or matrix–matrix product gates in total. This holds with arbitrary uniform entrywise functions and arbitrary two-input aggregation kernels in the model of Section 6.*

More precisely, writing $\#_{\mathrm{mv}}(C)$ and $\#_{\mathrm{mm}}(C)$ for the typed product counts just defined, every exact circuit satisfies

$$\beta\big(\#_{\mathrm{mv}}(C) + 3\#_{\mathrm{mm}}(C)\big) \geq 2^k.$$

For Boolean scalars the stronger inequality is $\#_{\mathrm{mv}}(C) + 2\#_{\mathrm{mm}}(C) \geq 2^k$. Each inequality holds separately at every $n \geq n_0(2^k)$. For an input-independent randomized circuit with pointwise error at most $\varepsilon < 1/2$, the corresponding assertions are

$$\beta\,\mathbb{E}\big(\#_{\mathrm{mv}}(C) + 3\#_{\mathrm{mm}}(C)\big) \geq (1-2\varepsilon)2^k,$$
$$\mathbb{E}\big(\#_{\mathrm{mv}}(C) + 2\#_{\mathrm{mm}}(C)\big) \geq (1-2\varepsilon)2^k \quad \text{(Boolean scalars)}.$$

For every fixed semiring in the theorem, the optimal total circuit size is $\Theta(2^k)$.

Proof. Use the $0_S, 1_S$ encodings of the Boolean G_k tests. Nontriviality makes these values distinct, and idempotence makes the value of $\mathcal{G}_k$ exactly its Boolean truth value under this encoding. Thus there are 2^k tests. Apply Theorem 3.4 and Lemma 6.3, or its Boolean refinement. Lemma 3.1 provides the construction at every stated size, and the expected-cost part of the theorem gives the randomized inequalities.

The upper bound uses exactly 2^k matrix–vector contractions. Define H_A by replacing the intersections in its earlier definition by pointwise products $\otimes$. For each A, contract H_{A^c} with the constant 1_S column vector, broadcast the result along rows, multiply entrywise by H_A, and join the outputs over A. Distributivity and commutativity prove that this computes $\mathcal{G}_k$ on arbitrary inputs. All H_A and H_{A^c} can be shared with $O(2^k)$ total nodes. Hence for a fixed semiring of finite breadth the minimum circuit size is $\Theta(2^k)$, and the positive ARA(3) source has linear size. □

In the Boolean one-sided model, mixing this exact computation with the constant-zero and constant-one outputs attains the randomized coefficient $1-2\varepsilon$, as in Section 4.3.

Examples of breadth-one semirings include max–min on $[0,1]$, $(\mathbb{R}\cup\{-\infty\}, \max, +, -\infty, 0)$, and $(\mathbb{R}\cup\{+\infty\}, \min, +, +\infty, 0)$. For each, at least $2^k/3$ data-dimension products are necessary, even in the enlarged kernel model. Propositions D.1 and D.2 in Appendix D give complementary bounds for the disjunctive query F_k over finite chains and tropical scalars.

6.4 Positive computations without idempotence

A separate transfer handles positive arithmetic, for which join breadth is not available. Let $(S, +, \otimes, 0_S, 1_S)$ be a nontrivial commutative semiring satisfying

$$r\,1_S \neq 0_S \qquad \text{for every integer } r \geq 1, \tag{NV}$$

where $r\,1_S$ is the sum of r copies of 1_S. Restrict the target to positive entrywise addition and multiplication, semiring matrix products, transpose, broadcasting, diagonal operations, and constants in $\mathbb{N}1_S = \{r\,1_S : r \geq 0\}$. Arbitrary entrywise decoding and negative constants are not allowed here. Replace joins in $\mathcal{G}_k$ by sums to define its positive semiring version.

On Boolean-encoded inputs, every intermediate belongs to $\mathbb{N}1_S$. The map sending 0_S to zero and every other element of this subsemiring to one is a homomorphism to the Boolean semiring. Indeed, sums and products of positive multiples of 1_S are again nonzero by the nonvanishing condition (NV). Applying this map to every gate of a target circuit C gives a Boolean matrix circuit with exactly the same typed product counts. Its output is G_k. The Boolean lower bound therefore gives $\#_{\mathrm{mv}}(C) + 2\#_{\mathrm{mm}}(C) \geq 2^k$. For a randomized circuit, correctness of a scalar output implies correctness after this map, so the same transfer gives $\mathbb{E}(\#_{\mathrm{mv}}(C) + 2\#_{\mathrm{mm}}(C)) \geq (1 - 2\varepsilon)2^k$. Here canonical cost is applied to the Booleanized computation; no idempotent transport argument is claimed for the original scalar circuit. The 2^k-product expansion still works by distributivity. This includes positive computations over $\mathbb{N}$ and nonnegative reals.

Appendix C.6 shows why the operation restrictions matter: arbitrary decoding together with nonidempotent summation can encode all row-profile counts in one aggregation. Conversely, the arbitrary-kernel result of Theorem 1.5 is valid because finite join breadth limits how many colors can support a single aggregation.

7 Open questions

Our bounds settle exponential FO^3-to-CoR compilation overhead over growing binary signatures. Several quantitative questions remain.

A fixed signature. Does a superpolynomial gap persist for full CoR over a fixed finite signature, or even one binary relation symbol? The encoding of [Nak22, Section 6] gives a superpolynomial separation for the existential fragment over a single symbol. Extending that conclusion to full CoR requires controlling unrestricted complement on the encoded inputs. Our present formulas use one or two binary symbols per coordinate.

The exact composition count for G_k. On arbitrary finite inputs the optimum lies between 2^{k-1} and 2^k; without converse it is exactly 2^k. Proposition C.1 attains 2^{k-1} on every input in the symmetric hub-recoloring family. Consequently that family alone cannot improve the full-CoR lower bound. Determining the optimum on arbitrary inputs calls for either a more efficient general circuit or a family exposing an additional obstruction.

Total size and tree size for F_k. Hansel's method determines the composition count up to a constant factor, but the optimum total circuit size remains between $\Omega(B_k)$ and $O(2^k)$. The tree-size

interval is $\Omega(kB_k)$ to $O(k2^k)$. Appendix C.4 proves that the present single-color incidence tests supply at most B_k tests. Closing these intervals may require an additional constraint on the Boolean work or a different choice of tests.

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A Explicit finite reference graphs

This appendix gives two deterministic versions of Lemma 3.1. The affine-plane construction follows the direction-partition method of Jipsen, Maddux, and Tuza [JMT95]. Their assignment of at least two directions to every color already guarantees two witnesses at plane order at least four; we include the verification because this is the property used here. The second construction retains the all-sufficiently-large-sizes guarantee of the random proof.

Proposition A.1 (Explicit reference graphs)**.** *For every palette size $N \geq 2$, a graph with the two-witness property of Lemma 3.1 can be constructed deterministically on fewer than $16N^2$ vertices. For every prescribed $n \geq n_0(N)$, such a graph on exactly n vertices can be constructed in time polynomial in n, N.*

We prove the two assertions in the following subsections.

A.1 A quadratic-size affine-plane construction

Let s be the smallest power of two with $s \geq 2N - 1$, and let $\mathbb{F}_s$ be the field of s elements. Use vertex set $V = \mathbb{F}_s^2$. There are $s+1$ directions of affine lines. Partition these directions into sets Dir_μ, $\mu \in \Lambda$, each of size at least two. Color distinct points u, v by the unique μ containing the direction of their joining line. Since $N \geq 2$, we have $s \geq 4$ and $|V| = s^2 < 16N^2$.

Fix distinct u, v and requested colors λ, μ. If $\lambda = \mu = \chi(u, v)$, all $s - 2 \geq 2$ other points of the line uv are witnesses. Otherwise, choose a direction from Dir_λ at u and a direction from Dir_μ at v, neither equal to the direction of uv, and choose the two directions to be distinct. Their two lines intersect at a unique witness outside $\{u, v\}$. There are at least two choices: if the colors differ, their direction sets are disjoint and at most one set loses a direction to uv; if they agree but differ from $\chi(u, v)$, two directions from their common set can be taken in either order. Distinct direction pairs give distinct intersection points, since a point outside $\{u, v\}$ determines both directions uniquely. Thus the coloring has the two-witness property. For $N = 2^k$, one can choose $s = 2N$, giving 4^{k+1} vertices. This construction supplies selected domain sizes, rather than every large size; induced subgraphs need not retain the extension property.

A.2 Every sufficiently large size

Fix any $n \geq n_0(N)$ and start with independently uniform edge colors. For distinct u, v, w and colors λ, μ, let $\mathcal{B}_{u,v,w,\lambda,\mu}$ be the event that there is no witness outside $\{u, v, w\}$. Their expected total number is less than one by the proof of Lemma 3.1. Under any partial assignment σ of edge colors, independence of the remaining edges gives

$$\Pr(\mathcal{B}_{u,v,w,\lambda,\mu} \mid \sigma) = \prod_{z \notin \{u,v,w\}} \left(1 - \Pr(\chi(u, z) = \lambda \mid \sigma) \Pr(\chi(v, z) = \mu \mid \sigma)\right).$$

Each factor uses probabilities in $\{0, 1/N, 1\}$. At the next uncolored edge, evaluate the sum of these conditional probabilities for each of its N possible colors and choose a color minimizing the sum. The average of the choices is the current sum, so this never increases it. After all edges are assigned, the number of bad events is an integer strictly below one, hence zero. This is a deterministic algorithm polynomial in n, N: there are polynomially many events, factors, edge choices, and bits in their

rational representations. The same argument applies to the directed construction in Appendix E.1, with independently colored ordered arcs.

B A fixed circuit for the Hansel bound

We prove Proposition 4.3. Hansel's monotone identification bound [Han66, AS17] concerns adaptive membership queries. Here the queried set may differ between rows, so we spell out the fixed wiring and its total number of Boolean gates. All notation in this appendix is local: Γ is a family of chains, $\gamma = (W_0 \subset \cdots \subset W_{|\gamma|-1})$ is one chain, and the auxiliary subsets V_i below are not vertex sets.

B.1 The symmetric chains

Identify a subset of $[k]$ with its bit string. Treat zero as an opening parenthesis and one as a closing parenthesis, and match them by the usual stack procedure. Unmatched positions have the pattern $1 \cdots 1 0 \cdots 0$. Hold all matched pairs fixed and let the unmatched positions run through all patterns of this form. This partitions the Boolean cube into saturated chains, each symmetric about rank $k/2$. Denote their family by Γ. Every chain meets rank $\lfloor k/2 \rfloor$ exactly once, so $|\Gamma| = B_k$. The number of chains of length at least two is $\binom{k}{\lfloor k/2 \rfloor + 1}$: for odd k every chain has even length; for even k, these are exactly the chains meeting the rank above the middle.

For $0 \le i \le |\gamma| - 3$ define

$$V_i = W_i \cup (W_{i+2} \setminus W_{i+1}).$$

Then $W_i \subset V_i \subset W_{i+2}$. The two newly affected unmatched positions form a new matched 01 pair; all previous pairs remain matched. Consequently V_i lies in a chain of length $|\gamma| - 2$. This is the chain property needed below. Process the chains in increasing order of length, so answers for every V_i have already been computed.

B.2 Two row queries per chain

For each subset $W \subseteq [k]$, let

$$E_W = R_W; \mathbf{1}.$$

Its value in a row says whether some entry in that row satisfies all relations with indices in W. These are row-constant relations, and $W \subseteq W'$ implies $E_{W'} \subseteq E_W$. For a chain γ, put $a_i = E_{V_i}$, and use Boolean operations on row-constant relations to define

$$l_j = \bigcup_{i \ge j} a_i, \qquad u_j = \bigcap_{i \le j-2} a_i, \qquad U_j = u_j \cap \overline{l_j} \quad (0 \le j < |\gamma|).$$

Indices i in these operations range from 0 to $|\gamma| - 3$; empty unions and intersections have their usual meanings. The chain property implies $l_j \subseteq E_{W_j} \subseteq u_j$. In any fixed row, the positions with $U_j = 1$ form an interval containing at most two adjacent positions. Indeed l_j and u_j are nonincreasing in j, and having both j and $j+2$ unknown would require $a_j = 0$ from $l_j = 0$ and $a_j = 1$ from $u_{j+2} = 1$.

Put $U_{-1} = U_{|\gamma|} = \mathbf{0}$, and define the two selectors

$$\mathrm{fst}_j = U_j \cap \overline{U_{j-1}}, \qquad \mathrm{lst}_j = U_j \cap \overline{U_{j+1}}.$$

At most one first and one last position is selected in each row. Compute

$$Z_{\mathrm{fst}} = \left(\bigcup_j (\mathrm{fst}_j \cap R_{W_j})\right); \mathbf{1}, \qquad Z_{\mathrm{lst}} = \left(\bigcup_j (\mathrm{lst}_j \cap R_{W_j})\right); \mathbf{1}.$$

Because the selectors depend only on the row, these products ask exactly the two selected membership queries, with no cross-talk between rows. Recover all answers by

$$E_{W_j} = l_j \cup (\mathrm{fst}_j \cap Z_{\mathrm{fst}}) \cup (\mathrm{lst}_j \cap Z_{\mathrm{lst}}).$$

For a singleton chain only the first query is needed. All other operations here are fixed Boolean gates; the circuit has no adaptive wiring.

Finally output

$$\bigcup_{W \subseteq [k]} (R_{W^c} \cap E_W),$$

where $W^c = [k] \setminus W$. This is the expansion of F_k. The number of row projections is at most

$$B_k + \binom{k}{\lfloor k/2 \rfloor + 1} = B_{k+1}.$$

Precompute all R_W by sharing intersections along a subset tree. Within a chain, suffix unions and prefix intersections compute all l_j, u_j with $O(|\gamma|)$ nodes, and the selectors, queries, and reconstruction take another $O(|\gamma|)$ nodes. Since $\sum_{\gamma \in \Gamma} |\gamma| = 2^k$, the total circuit size is $O(2^k)$. A query for a chain depends only on chains at least two shorter, so any path crosses at most $\lfloor k/2 \rfloor + 1$ projection gates.

Replacing row projections by rebinding y also gives a FO^2 formula circuit with at most B_{k+1} distinct quantifier gates and $O(2^k)$ total nodes. Expanding the shared definitions may duplicate queries; this is not an upper bound of B_{k+1} on quantifier occurrences of an ordinary formula, or on composition occurrences of a CoR term.

C Tree bounds and limits of the method

C.1 The input-occurrence charging lemma

This subsection proves the tree refinement stated in Section 5.2.

Lemma 5.2 (Disjoint input-occurrence charging)**.** *Fix the reference structure and one-color folds of Section 3.4. Suppose every singleton selector of a target color requires at least $d > 0$ input-relation occurrences. Then every* CoR *term T, whether correct or not, satisfies*

$$\mathrm{cost}(T) \leq \#_{\mathrm{rel}}(T)/d.$$

Proof. For a reference subterm X, let $S_X \subseteq \Lambda$ be its off-diagonal support. Boolean operations act on these sets in the usual way and converse fixes them. By the color-table identities (CT), a composition $X;Y$ with both supports nonempty has off-diagonal support Λ. Call such a node *active*. If either support is empty, the off-diagonal profile of the product is zero or one of the two operand profiles, according to the diagonal bits. Such an inactive node has cost zero: every generic join in the local calculation vanishes.

Construct a Boolean formula for the off-diagonal profile of any subterm by replacing active compositions by the constant true, inactive compositions by zero or the relevant operand profile, and converse by its argument. This operation never duplicates an input occurrence. For each active node $g = (X; Y)$, apply this pruning to its two operands. Let $\mathcal{L}_g$ be the set of original input occurrences that remain in the two resulting profile formulas. These are the occurrences that *survive the pruning*, not all input occurrences below g. This distinction prevents repeated charging at successive active nodes. The sets $\mathcal{L}_g$ are pairwise disjoint: an active descendant is replaced by a constant when an ancestor's operands are pruned, and incomparable nodes have disjoint subtrees.

For a fold deleting target color a, the local table shows that a composition can fail to commute with transport only if

$$S_X = \{a\}, \qquad S_Y = \{a\}, \qquad \text{or} \qquad S_X \cap S_Y = \{a\}.$$

Let E_g be the set of targets satisfying at least one of these conditions. Thus $c(g) \le |E_g|$. If $S_X \cap S_Y$ is nonempty, E_g has size at most one. Its unique target, if present, is selected by one operand profile or by their conjunction. The selector assumption therefore gives $|\mathcal{L}_g| \ge d|E_g|$. If the two supports are disjoint, only singleton operands contribute to E_g. Each contributing operand has at least d input occurrences, and the operands are disjoint trees. The same inequality follows. Summing over active nodes proves

$$\mathrm{cost}(T) = \sum_g c(g) \le \sum_g |E_g| \le \frac{1}{d} \sum_g |\mathcal{L}_g| \le \#_{\mathrm{rel}}(T)/d.$$

Diagonal bits may affect which inactive profile is retained, but do not create additional input occurrences in this argument. □

C.2 Factored trees for G_k

A complete binary decision tree successively chooses P_i or Q_i for the read edge. At leaf A place $H_{A^c}; \mathbf{1}$. A branch at coordinate i with child terms U_1, U_0 has the form

$$(P_i \cap U_1) \cup (Q_i \cap U_0).$$

Distributing the branches recovers the DNF for G_k on arbitrary inputs; no one-hot assumption is used. Each of the 2^k leaves costs $2k + 1$ nodes, and each of the $2^k - 1$ internal branches adds five. The total is $(2k + 6)2^k - 5$. The number of relation-input occurrences is $k2^k + 2(2^k - 1)$. Replacing each leaf by $\exists y\, H_{A^c}(x, y)$ gives a FO^2 tree with $(2k + 5)2^k - 5$ nodes.

C.3 Why the profiles are crossed

For comparison, the superficially similar non-crossed formula collapses:

$$\exists z \bigwedge_{i=1}^{k} [(P_i(x, y) \wedge P_i(x, z)) \vee (Q_i(x, y) \wedge Q_i(x, z))] \equiv \bigwedge_{i=1}^{k} (P_i(x, y) \vee Q_i(x, y)).$$

The forward implication is immediate; for the reverse implication take $z = y$. Crossing the two profiles in G_k prevents this simplification.

C.4 The antichain limit for Boolean F_k

In the incidence-color interpretation, a read profile A^c accepts every witness color containing A. To distinguish the removal of a single accepted color, that color must be the unique accepted palette color. In particular, all such target colors are inclusion-maximal palette colors and form an antichain. An antichain $\mathcal{H} \subseteq 2^{[k]}$ has size at most B_k: a uniformly random ordering of $[k]$ has a given A as an initial set with probability $1/\binom{k}{|A|}$, and at most one member of an antichain can occur among its initial sets. Thus $\sum_{A\in\mathcal{H}} \binom{k}{|A|}^{-1} \le 1$ and $|\mathcal{H}| \le B_k$. This is a limit of single-color deletion tests with incidence profiles, not an upper bound on every possible lower-bound method for F_k.

C.5 Converse on the promised symmetric family

Proposition C.1 (Sharpness on the symmetric recoloring family)**.** *For every $k \ge 1$, on the cube reference structure and all its hub-star recolorings by maps $f : \Lambda \to \Lambda$, the minimum number of composition gates in a full-*CoR *circuit computing G_k is 2^{k-1}.*

Proof. Let $N = 2^k$ and retain the symmetric cube reference structure. For each unordered complementary pair $\{A, A^c\}$ choose one representative A and compute, once,

$$X_A = H_A; H_{A^c}.$$

Use the shared result in

$$\bigcup_{\{A,A^c\}} \left[(H_{A^c} \cap X_A) \cup (H_A \cap X_A^{\smile})\right].$$

This circuit has exactly $N/2$ compositions and computes G_k on every structure obtained by recoloring the entire star of the fixed hub by an arbitrary map f. To check this, write $J = f(\Lambda)$. The relation X_A is true on all distinct non-hub pairs, has zero diagonal, and has hub-row and hub-column values respectively $[A \in J]$ and $[A^c \in J]$. The masks select the complementary read colors. At a non-hub vertex every color still occurs, so G_k is true on every off-diagonal read edge there. At the hub it is true precisely when the complementary color lies in J. These are exactly the displayed circuit's values. The lower bound follows by applying Theorem 3.4 and Lemma 3.5 to the N one-color folds in this family. Thus the circuit attains the optimum on the promised family. □

C.6 Scalar packing and the role of the operation model

The Boolean promised-family example shows why a composition can require a charge of two. The next example explains the three scalar observations at the hub. It concerns the same promise, not an upper bound on arbitrary input structures.

Proposition C.2 (Three scalar observations)**.** *Let $N = 2^k$, with $k \ge 1$. On the symmetric cube reference structure and all its hub-star recolorings by arbitrary maps $f : \Lambda \to \Lambda$, the Boolean query G_k can be computed with at most $\lceil N/3 \rceil + 1$ max–min products over $[0, 2]$, using arbitrary uniform entrywise functions, transpose, and diagonal operations.*

Proof. Put $m = \lceil N/3 \rceil$. Choose m distinct complementary pairs $\{A_i, A_i^c\}$, and assign each of the remaining $r = N - 2m \le m$ colors to a different pair. These choices are possible for $N = 2^k \ge 2$.

Call an assigned residual color ρ_i. Input profile tests and entrywise functions construct arrays X_i, Y_i whose off-diagonal values (p_λ, q_λ) are

$$(2,0) \text{ at } A_i, \quad (0,2) \text{ at } A_i^c, \quad (1,1) \text{ at } \rho_i \text{ if assigned}, \quad (0,0) \text{ otherwise.}$$

Put both diagonal values zero and compute one max–min product S_i. Write $J = f(\Lambda)$ for the colors still present at the hub. The generic value of S_i at distinct non-hub vertices is two. At the hub, the local product calculation gives

$$[S_i(h,v) = 2] = [A_i \in J], \qquad [S_i(u,h) = 2] = [A_i^c \in J], \qquad S_i(h,h) = [\rho_i \in J]$$

for non-hub u, v, with the last equality only for assigned pairs. The diagonal value at any non-hub vertex is one for such a pair. Read masks $H_{A_i^c}, H_{A_i}$, together with S_i and its transpose, therefore recover the answers for both selected colors on every non-loop read edge. All loops are suppressed by these masks.

It remains to recover the residual colors. For each assigned ρ_i, diagonal extraction followed by an entrywise test gives a vector

$$b_i(u) = [S_i(u,u) < 1].$$

It is zero outside h, and at h it records that ρ_i is missing. Encode these finitely many bits in distinct binary positions of a single nonnegative scalar, rescaled into $[0,2]$, at each diagonal entry of an array. One row projection, a max–min product with the constant-two array, broadcasts this code along its row. A fixed entrywise decoder recovers each bit. Intersecting its complement with the read mask $H_{\rho_i^c}$ supplies exactly the answers for witness color ρ_i. At non-hub vertices every color is present and all decoded missing bits are zero, as required. Combining these outputs with those for the selected pairs computes G_k on the entire promised family. The construction uses m products S_i and at most one final projection. □

For a b-bit state with coordinatewise disjunction as join, b tests can instead be packed into one row projection. On arbitrary Boolean inputs, group the 2^k DNF witness tests into blocks of at most b, aggregate their truth values in parallel, and decode and mask the resulting bits entrywise. This uses $\lceil 2^k/b \rceil$ projections. It complements the lower bound $b(\#_{\mathrm{mv}}(C) + 2\#_{\mathrm{mm}}(C)) \geq 2^k$ of Section 6.

Finite join breadth, or a positive-semiring restriction, cannot simply be removed. With ordinary real addition and arbitrary entrywise decoding, all finite row histograms of Boolean input profiles can be encoded in one projection: for an n-element domain, encode profile number j by $(n+1)^j$ and sum along the row. Base-$(n+1)$ digits recover every profile count. A read-edge-dependent decoder then evaluates F_k or G_k. This example allows arbitrary nonpositive entrywise functions and thus lies outside the positive fragment of Section 6.4.

D Weighted disjunctive queries

This appendix proves Propositions D.1 and D.2, supplementary applications of the common test–cost theorem to finite-chain and tropical versions of F_k. The notation in this appendix is local: ℓ is the number of positive levels in a chain, and $t_{k,\ell}, \widehat{t}_{k,\ell}$ are the lower- and upper-bound counts defined below. Throughout $k \geq 2$.

D.1 A finite-chain transition

On $S = \{0, 1, \dots, \ell\}$, with $\ell \geq 1$, use maximum as addition and minimum as multiplication, and define

$$\mathcal{F}_{k,\ell}(x, y) = \max_{z} \min_{i \in [k]} \max(R_i(x, y), R_i(x, z)).$$

Put

$$t_{k,\ell} = \sum_{j=\lceil k/2 \rceil}^{\min(k, \lceil k/2 \rceil + \ell - 1)} \binom{k}{j},$$
$$\widehat{t}_{k,\ell} = \sum_{j=\max(0, \lfloor k/2 \rfloor - \ell + 1)}^{\min(k, \lfloor k/2 \rfloor + \ell)} \binom{k}{j}.$$

Symmetry and unimodality of binomial coefficients give $\widehat{t}_{k,\ell} \leq 2t_{k,\ell}$. They also give

$$t_{k,\ell} = \Theta\Big(\min\{2^k, \ell B_k\}\Big),$$

with absolute constants: the upper bound follows by counting levels and all subsets, and the first $\min\{\ell, \lfloor \sqrt{k} \rfloor\}$ levels above the middle each have size at least a constant multiple of B_k. For small k the constants can be adjusted uniformly.

Proposition D.1 (Finite-chain disjunctive queries)**.** *For $k \geq 2$ and $\ell \geq 1$, the minimum number of matrix–vector, vector–matrix, and square matrix–matrix products needed to compute $\mathcal{F}_{k,\ell}$ on all finite inputs is $\Theta(t_{k,\ell})$, with absolute constants. This holds in the enlarged max-aggregation model with arbitrary uniform entrywise functions and two-input kernels. More precisely, every correct circuit satisfies $\#_{\mathrm{mv}}(C) + 3\#_{\mathrm{mm}}(C) \geq t_{k,\ell}$. There is an exact circuit using at most $\widehat{t}_{k,\ell} \leq 2t_{k,\ell}$ row projections.*

Proof. Let $\mathcal{H}$ consist of the ℓ upper levels in the definition of $t_{k,\ell}$, truncated at k, and let $\Lambda = \mathcal{H} \cup \{A^c : A \in \mathcal{H}\}$. For a color $\mu \in \Lambda$, assign the amplitude

$$a(\mu) = \begin{cases} \ell - (|\mu| - \lceil k/2 \rceil), & \mu \in \mathcal{H}, \\ \ell, & \mu \notin \mathcal{H}. \end{cases}$$

Set $R_i(\mu) = a(\mu)$ when $i \in \mu$ and zero otherwise, and put all loop values zero. For a target $A \in \mathcal{H}$, use read color A^c and fold A to A^c. The read values are ℓ on A^c and zero on A. A witness color contributes zero unless it contains A. Color A contributes $a(A) > 0$, and every strict superset in the palette contributes a smaller amplitude. Loops contribute zero. Thus the value strictly decreases when color A disappears from the hub row. There are $t_{k,\ell}$ tests. Theorem 3.4 gives

$$\#_{\mathrm{mv}}(C) + 3\#_{\mathrm{mm}}(C) \geq t_{k,\ell}$$

for the enlarged max-aggregation model with arbitrary kernels, and $(1 - 2\varepsilon)t_{k,\ell}$ on the right for expected randomized counts. For $\ell = 1$, the Boolean bound improves the coefficient three to two. For a fixed domain size it suffices that $n \geq n_0(|\Lambda|)$.

For completeness, a finite-valued version of the circuit in Appendix B uses at most $\widehat{t}_{k,\ell}$ row projections. Now put $R_W = \min_{i\in W} R_i$, with $R_\varnothing = \ell$, and $E_W(x) = \max_z R_W(x,z)$. Use the same chains and auxiliary sets V_i. With $a_i = E_{V_i}$, the bounds

$$l_j = \max_{i\geq j} a_i \leq E_{W_j} \leq u_j = \min_{i\leq j-2} a_i$$

hold, with empty maximum zero and empty minimum ℓ. At each threshold $\eta \in \{1,\dots,\ell\}$, the Boolean proof shows at most two unknown positions. Hence at most 2ℓ positions satisfy $l_j < u_j$ in any row. Put $s_\gamma = \min\{|\gamma|, 2\ell\}$. For $1 \leq r \leq s_\gamma$, let the row-constant selector $\sigma_{r,j}(x)$ be ℓ if j is the rth unknown position in row x, and zero otherwise. A fixed prefix computation implements these selectors. Compute one row projection for each r:

$$Z_r(x) = \max_z \max_{0\leq j<|\gamma|} \min\{\sigma_{r,j}(x), R_{W_j}(x,z)\}.$$

In each row at most one j is selected, so $Z_r(x)$ is exactly its membership-query answer, or zero when that row has no rth unknown position. Recover every answer by

$$E_{W_j}(x) = \max\left\{ l_j(x),\ \max_{1\leq r\leq s_\gamma} \min\{\sigma_{r,j}(x), Z_r(x)\} \right\}.$$

For an unknown position the selected answer lies between l_j and u_j; for a known position all selectors vanish and $l_j = u_j = E_{W_j}$. A prefix computation of the number of unknown positions, using one-hot Boolean states up to $\min\{|\gamma|, 2\ell\}$, implements all these selectors with $O(\min\{\ell,k\}|\gamma|)$ gates. No data-dependent wiring is required.

A chain needs at most $\min\{|\gamma|, 2\ell\}$ queries. By symmetry this is its number of elements in the central 2ℓ ranks used to define $\widehat{t}_{k,\ell}$. Summing over chains gives the stated query count, $O(\min\{\ell,k\}2^k)$ total nodes, and composition depth at most $\lfloor k/2 \rfloor + 1$. The output is $\max_{W\subseteq[k]} \min(R_{W^c}, E_W)$, equal to $\mathcal{F}_{k,\ell}$ by distributivity. The direct expansion alternatively uses at most $2^k - 1$ projections and $O(2^k)$ shared nodes. Consequently the optimal number of data-dimension products is $\Theta(t_{k,\ell})$. When $\ell = \Omega(\sqrt{k})$, total circuit size is $\Theta(2^k)$ as well. For smaller ℓ, no matching total-size bound is implied by the product count alone.

□

D.2 A tropical disjunctive query

Over max–plus define

$$\mathcal{F}_k^{\max,+}(x,y) = \max_z \sum_{i=1}^{k} \max(R_i(x,y), R_i(x,z)), \qquad S = \mathbb{R} \cup \{-\infty\}.$$

Proposition D.2 (Tropical disjunctive queries)**.** *For $k \geq 2$, the minimum product count and the minimum total circuit size for computing $\mathcal{F}_k^{\max,+}$ on all finite inputs are both $\Theta(2^k)$, even with arbitrary uniform entrywise functions and two-input max-aggregation kernels. The corresponding statement holds over min–plus after negating finite inputs and exchanging the two infinities.*

Proof. Use the full cube $\Lambda = 2^{[k]}$, and assign

$$R_i(\mu) = \begin{cases} 1/(|\mu|+1), & i \in \mu, \\ 0, & i \notin \mu, \end{cases} \qquad R_i(u,u) = -\infty.$$

Targets are the colors A with $|A| \geq \lceil k/2 \rceil$, and $t_k^{\max,+} = \sum_{j=\lceil k/2 \rceil}^{k} \binom{k}{j} \geq 2^{k-1}$. Read at A^c and fold A to A^c as before. Subtract the fixed read-edge baseline from a witness score. A color μ then contributes

$$\Delta_A(\mu) = \frac{|A \cap \mu|}{|\mu|+1} + |A^c \cap \mu| \max\left\{ \frac{1}{|\mu|+1} - \frac{1}{|A^c|+1}, 0 \right\}.$$

If $|\mu| < |A|$, this is at most $|\mu|/(|\mu|+1)$, strictly below $|A|/(|A|+1)$. If $|\mu| \geq |A| \geq |A^c|$, the second term is zero; the first attains $|A|/(|A|+1)$ only for $\mu = A$. Loop witnesses add zero to the baseline. Therefore A is the unique maximizing color, and its fold strictly decreases the answer. The input uses only $O(\log k)$-bit rational amplitudes and $-\infty$.

The same theorem gives $\#_{\mathrm{mv}}(C) + 3\#_{\mathrm{mm}}(C) \geq t_k^{\max,+}$, including arbitrary max-aggregation kernels and arbitrary uniform entrywise functions. Distributing the maxima gives an upper bound of $2^k - 1$ row projections and $O(2^k)$ total nodes, since all subset sums can be shared. Thus total circuit size and product count are $\Theta(2^k)$. Randomization multiplies the lower count by $1 - 2\varepsilon$. Negating all finite scalar values and exchanging $-\infty$ with $+\infty$ gives the corresponding min–plus statement. These are exact-value lower bounds, not bounds for numerical approximation. □

E Directed targets and two-edge witness queries

We give a sharper bound without converse, then independently analyze the two-edge witness queries Ψ_k, Φ_k using anchors. All counting bounds below follow from Theorem 3.4, including its expected-cost extension with factor $1 - 2\varepsilon$.

E.1 A directed construction without converse

Corollary 4.4 (Without converse)**.** *Every converse-free relational circuit defining F_k on finite structures has at least B_k composition gates, for $k \geq 2$. Every such circuit defining G_k has at least 2^k composition gates, for $k \geq 1$. The latter bound is exact.*

Proof. Color each ordered pair of distinct vertices independently from a palette Λ; write $\chi(u,v)$ for its color, with no relation between $\chi(u,v)$ and $\chi(v,u)$. Require that for every u, v, now allowing $u = v$, and every $\lambda, \mu \in \Lambda$, at least two vertices outside $\{u, v\}$ realize

$$(\chi(u,z), \chi(z,v)) = (\lambda, \mu).$$

Candidate witnesses use disjoint ordered arcs, even when $u = v$. Lemma 3.1 and its deterministic construction therefore apply with the same threshold $n_0(N)$ and at most n^3N^2 bad events.

Reference arrays have a diagonal value and one value per forward color. These tables are closed under entrywise operations and aggregation. Every output pair, including a diagonal, has generic contribution $\kappa_* = \bigvee_{\lambda,\mu} \kappa(p_\lambda, q_\mu)$; on a diagonal the sole endpoint contribution is $\kappa(p_{\mathbf{id}}, q_{\mathbf{id}})$. Recolor only the *outgoing* arcs at h by a map f, leaving all incoming arcs unchanged. Transport uses this

new forward coloring. Every output pair outside row h is unchanged after transport. On row h, including (h, h), the only altered generic contribution is

$$\bigvee_{\lambda \in f(\Lambda),\, \mu \in \Lambda} \kappa(p_\lambda, q_\mu).$$

All endpoint contributions already have the transported values. At most β first colors support this one join. Hence every aggregation has cost at most β, and every Boolean composition has cost at most one.

Use the same palettes, input incidence profiles, read edges, and folds as for F_k, G_k. These queries inspect outgoing witness edges at h, so their tests are unchanged. Theorem 3.4 gives

$$g_{;}(C) \geq B_k \quad (C \equiv_{\text{fin}} F_k), \qquad g_{;}(C) \geq 2^k \quad (C \equiv_{\text{fin}} G_k)$$

for converse-free relational circuits. The G_k bound is attained by its DNF on arbitrary inputs. The scalar bound also applies to the weighted F tests of Appendix D. Converse is excluded because a forward color table does not determine the reverse color; the coefficient-one argument consequently does not apply to full CoR.

□

E.2 The two-edge source formulas

The two-edge queries are

$$\Psi_k(x, y) = \exists z \bigwedge_{i=1}^{k} (R_i(x, z) \vee R_i(z, y)) \qquad (k \geq 2),$$

$$\Phi_k(x, y) = \exists z \bigwedge_{i=1}^{k} [(P_i(x, z) \wedge P_i(z, y)) \vee (Q_i(x, z) \wedge Q_i(z, y))] \qquad (k \geq 1).$$

The first is the source family used in [Nak22] for positive CoR. Their direct expansions on arbitrary inputs are

$$\Psi_k \equiv \bigcup_{I \subseteq [k]} (R_I; R_{I^c}), \qquad \Phi_k \equiv \bigcup_{A \subseteq [k]} (H_A; H_A).$$

Each uses 2^k compositions. Both witness edges pass through the quantified variable, unlike in F_k, G_k.

Theorem E.1 (Two-edge witness queries)**.** *Every full-*CoR *relational circuit defining Ψ_k on all finite structures has at least $\lceil B_k/2 \rceil$ composition gates, for $k \geq 2$. Every such circuit defining Φ_k has at least 2^{k-1} composition gates, for $k \geq 1$.*

The following construction and transport calculation prove the theorem.

Take the symmetric reference core V with the middle-layer palette for Ψ_k, or the full cube for Φ_k. Add a disjoint set of vertices $\mathcal{A} = \{\alpha_A : A \in \mathcal{H}\}$, called anchors. For Ψ_k, set the input profile of every edge (α_A, u), $u \in V$, to A^c: $R_i(\alpha_A, u) = 1$ exactly when $i \in A^c$. On core edges use the original incidence profiles. For Φ_k, use profile A at (α_A, u): $P_i = 1$ exactly for $i \in A$, and $Q_i = 1$ exactly for $i \in A^c$; core edges use the one-hot cube interpretation. All core-to-anchor, anchor-to-anchor,

and loop input entries are zero. Fold core color A to A^c on the symmetric hub star, leaving every anchor edge unchanged, and read the output at (α_A, h). A core witness is accepted exactly when its edge to h has color A. The witness h fails because its loop inputs are zero. Anchor witnesses also fail: for Ψ_k their outgoing profile omits a nonempty set, and for Φ_k the first witness edge has all-zero profile. Hence the reference answer is one and the folded answer is zero. There are B_k and 2^k tests, respectively.

E.3 Transport in the presence of anchors

A reference intermediate relation has one value on the core diagonal, one on each core color, one on each anchor-to-core and core-to-anchor class, and one on each ordered pair of anchors. Define transport by recoloring the core entries and leaving all anchor classes fixed. These classes are closed under Boolean operations, converse, and composition: generic core witnesses supply all color pairs for distinct core endpoints, all equal-color profiles (λ, λ) on a core diagonal, all colors for an anchor/core pair, and a constant value for an anchor/anchor pair. Anchor witnesses contribute fixed values.

For a composition with core profiles p, q, retaining the three joins

$$\bigvee_\lambda p_\lambda, \qquad \bigvee_\lambda q_\lambda, \qquad \bigvee_\lambda (p_\lambda \wedge q_\lambda)$$

is sufficient. On core/core outputs this is precisely the local calculation of Section 3.3, with fixed anchor contributions added. On an output (α, h), only a fixed anchor-to-core value conjoined with the second join can change; on (h, α), only the analogous first join can change. For non-hub core endpoints the two-witness property leaves all colors available outside h. On anchor/anchor outputs all transported operands are unchanged. Thus the two-color proof of Lemma 3.3 still applies. Read observations lie in unchanged anchor classes, so the observation condition (O) holds.

Applying Theorem 3.4 proves Theorem E.1. Since $|T| \geq 2\#_;(T) + 1$, the corresponding term-size lower bounds are $2\lceil B_k/2 \rceil + 1$ and $2^k + 1$. The composition bounds hold separately at every total size at least $n_0(N) + |\mathcal{H}|$, with the palette and target set used above. For an input-independent distribution with pointwise error at most $\varepsilon < 1/2$, the corresponding expected composition counts are at least $(1 - 2\varepsilon)B_k/2$ for Ψ_k and $(1 - 2\varepsilon)2^{k-1}$ for Φ_k. The input-occurrence refinement of Section 5.2 is specific to the unanchored construction.

The anchors provide the unchanged observations for these two-edge queries. On the same interpretations without anchors, the tests disappear. On the middle-layer recolored cores, $\Psi_k = \overline{\mathbf{id}}$. For Φ_k put $E = \bigcap_{i=1}^k (P_i \cup Q_i)$; on the cube cores, $E = \overline{\mathbf{id}}$ and $\Phi_k = E; E = \mathbf{1}$. At distinct endpoints the surviving generic profiles supply complementary colors for Ψ_k and equal colors for Φ_k; on a diagonal, symmetry prevents the former and permits the latter. Thus neither query detects a hub fold on these unanchored families.

E.4 A weighted two-edge query and its scope

Remark E.2 (Weighted two-edge queries). Over a nontrivial commutative idempotent semiring of breadth β, write the weighted two-edge query as

$$\Phi_k^{\mathrm{w}}(x, y) = \bigvee_{z \in \mathrm{dom}(\mathcal{M})} \bigotimes_{i=1}^{k} [(P_i(x, z) \otimes P_i(z, y)) \vee (Q_i(x, z) \otimes Q_i(z, y))].$$

The $0_S, 1_S$ anchor construction again gives 2^k tests. For ordinary semiring products, it suffices to retain the two core marginals $\bigvee_\lambda p_\lambda$, $\bigvee_\lambda q_\lambda$ and the core diagonal join $\bigvee_\lambda (p_\lambda \otimes q_\lambda)$. Distributivity reduces every anchor/core response to a fixed anchor coefficient multiplied by a marginal. At most 3β colors support these three joins, even if arbitrary uniform entrywise functions produce the intermediate operands. Consequently a circuit needs at least $2^k/(3\beta)$ semiring data-dimension products. Typed vector contractions obey the same upper charge, and inner products have zero charge, since reference vectors are constant on the core and transport fixes anchor coordinates. The expansion $\bigvee_A (H_A; H_A)$ gives the 2^k-product upper bound. The positive-semiring transfer of Section 6.4 also applies under its stated restrictions.

Arbitrary kernels cannot be substituted for multiplication in this last claim. On Boolean inputs, encode at each edge the set of profiles A for which H_A holds as one nonnegative real number. A two-input kernel returns one exactly when the two encoded sets intersect. One max-aggregation then computes Φ_k on every input. For Ψ_k, encode the set of true R_i and let the kernel test whether the union of its two decoded sets is $[k]$. This also uses one aggregation. Thus finite breadth alone is insufficient for the anchored two-edge queries with arbitrary kernels. The kernel lower bound for the read-edge/witness-edge query $\mathcal{G}_k$ in Section 6.3 has a different scope and remains valid.

F The classical single-exponential translation

We include the size analysis for the classical expressive equivalence [TG87, Mad06]. The relevant point is that successive quantifier eliminations need not create a tower of exponentials: an already translated subformula is treated as one leaf of the next Boolean expansion, without expanding its algebraic representation again.

Proposition 1.2 (Classical translation)**.** *Every* FO^3 *formula* φ *over binary relations with at most two free variables has an equivalent* CoR *term of size* $2^{O(|\varphi|)}$ *on all nonempty structures. Here size counts syntax-tree nodes.*

Proof. We represent a formula with at most two free variables as a binary relation of assignments to an ordered pair containing those variables. An unused coordinate is a dummy coordinate: membership is independent of it. Thus a unary formula is represented by a relation constant along rows or columns, and a sentence by $\mathbf{0}$ or $\mathbf{1}$. If a representation is moved to another ordered pair containing the same free variables, it is unchanged or replaced by its converse. This also handles the placement of a unary free variable in either coordinate. Closed formulas need no adjustment.

An atom with two distinct variables is represented by its input relation, its converse, or $\mathbf{id}$, as appropriate. A reflexive atom $R(u, u)$, when u is the first coordinate, is represented by $(R \cap \mathbf{id}); \mathbf{1}$; when it is the second coordinate, use the converse. An equality $u = u$ is represented by $\mathbf{1}$. These translations have constant size. Boolean operations on formulas over a fixed pair become intersection, union, and complement.

Process quantified subformulas from the inside out. Consider $\exists w\, \psi$, and let u, v be the other two variable names, in that order. Its free variables are contained in $\{u, v\}$. In the body ψ, regard each atom and each maximal quantified subformula as one leaf of a Boolean formula, called its *shell*. Each leaf has at most two free variables: this is immediate for binary atoms, and a quantifier binds one of the three available names. The quantified leaves have already been translated. Crucially, the shell expansion treats each of their translations as an opaque term.

If the shell has r leaf occurrences, a truth-table DNF has at most 2^r clauses, each using at most r literals. Repeated or logically dependent leaves cause no problem; the propositional identity remains valid after substituting their actual interpretations. In each clause, assign its literals to three groups according to the free-variable pair: (u, v), (u, w), or (w, v). A literal with only one free variable is assigned to any group containing it; a closed literal can be put in the first group. Use converse to orient each translated literal for its chosen pair, and use complement for a negative literal. Writing the conjunctions of the groups as $A(u, v)$, $B(u, w)$, and $C(w, v)$, the clause translates as

$$\exists w\,(A(u, v) \wedge B(u, w) \wedge C(w, v)) \quad \longmapsto \quad A \cap (B; C).$$

An empty group is represented by $\mathbf{1}$. This identity also covers vacuous quantification because domains are nonempty. Take the union of the translated clauses. Universal quantification uses $\forall w\,\psi \equiv \neg\exists w\,\neg\psi$, which introduces no subformula duplication. At the root, any remaining Boolean shell with at most two free variables is translated directly, after orienting its leaves to a common ordered pair. This proves correctness on all nonempty structures by induction.

For the size bound, cut the original syntax tree at quantifiers into the shells just used. Count each child quantified subformula as one leaf of its parent's shell. The sum of the shell sizes, including these boundary leaves, is $O(|\varphi|)$: each original node is counted once, with only one extra boundary occurrence per quantified subformula. A shell of size s creates at most $2^{O(s)}$ local nodes and at most $2^{O(s)}$ copies of each child translation. Along any path from an original leaf or a newly introduced local node to the final output, the resulting duplication factor is therefore at most

$$\prod_{\text{shells on the path}} 2^{O(s)} \leq 2^{O(|\varphi|)}.$$

There are $O(|\varphi|)$ original nodes and shell locations, and the local exponential creation factor is included in this same bound. Summing their contributions gives $|\varphi|\, 2^{O(|\varphi|)} = 2^{O(|\varphi|)}$ nodes. No step expands the internal Boolean structure of a translated child, which is what prevents an iterated-exponential recurrence. □